

Magnetic and Superconducting Materials

3.1 INTRODUCTION

The materials which can be made to behave like a magnet and which are easily magnetic field called as a magnetic materials.

3.1.1 Basic Definitions

1. Magnetic Dipole Moment (M)

The dipole moment is defined as the product of magnetic pole strength and length of the magnet. It is given by $M = ml$. Amp m^2 .

2. Magnetic Field

The space around which the magnetic lines of forces exist is called as magnetic field. Magnetic field is produced by permanent magnets such as a horse shoe magnet and temporarily by electromagnets (or) superconducting magnets.

3. Magnetic Lines of Force

The continuous curve in a magnetic field that exists from north pole to south pole is called as magnetic lines of force.

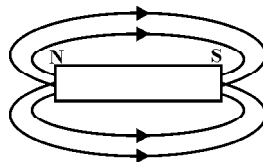


Fig. 3.1 Magnetic lines of force

4. Magnetic Induction (B) (or) Magnetic Flux Density

It is the number of magnetic lines of force passing through unit area of cross section.

$$B = \frac{\phi}{A} \text{ Wb m}^{-2} \text{ (or) Tesla}$$

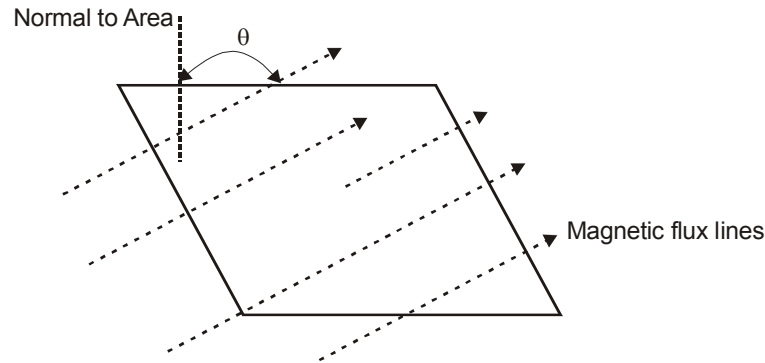


Fig. 3.2 Magnetic induction

5. Magnetic Field Strength (or) Magnetizing Field (H)

It is the force experienced by a unit north pole placed at a given point in a magnetic field. The magnetic induction B due to the magnetic field of intensity H applied in vacuum is related by $B = \mu_0 H \text{ Amp m}^{-1}$.

6. Magnetic Flux (ϕ)

The total number of magnetic lines of force passing through a surface.

Unit : Weber.

7. Intensity of Magnetization (I)

Magnetization is the process of converting a non-magnetic material into a magnetic material.

It is also defined as the magnetic moment per unit volume. $I = m / V \text{ web m}^{-2}$.

8. Magnetic Permeability (μ)

It is ratio of the magnetic induction (B) to the applied magnetic field intensity (H).
 $\mu = B / H$

Unit: Henry m^{-1}

It is the measure of ability of the material to permit magnetic lines of force.

9. Relative Permeability (μ_r)

It is defined as the ratio of permeability of the medium to the permeability of the free space (μ_r). $\mu_r = \mu / \mu_0$.

10. Magnetic Susceptibility (χ)

It is defined as the ratio of intensity of magnetization (I) and intensity of magnetic field (H). $\chi = I / H$.

The sign and magnitude of χ are used to determine the nature of the magnetic materials.

11. Bohr Magnetron (μ_B)

The orbital magnetic moment and the spin magnetic moment of an electron in an atom can be expressed in terms of atomic unit of magnetic moment called as Bohr magnetron.

$$1 \text{ Bohr magneton} = \frac{e\hbar}{2m} \Rightarrow \mu_B = 9.27 \times 10^{-24} \text{ Am}^2$$

12. Relation between susceptibility (χ) and Relative permeability (μ_r)

We know that when a current is supplied through a coil, magnetic field is developed. When a magnetic material is placed inside a external magnetic field, the magnetic flux density (B) arises due to applied magnetic field (H) and also due to the induced magnetization (I).

i.e., the total flux density,

$$B = \mu_0 (I + H) \quad \dots (1)$$

$$\text{But we know that } \mu = B / H \quad \dots (2)$$

From equations (1) and (2) we get,

$$\mu H = \mu_0 (I + H)$$

$$\frac{\mu}{\mu_0} = \frac{(I + H)}{H}$$

$$\mu_r = 1 + \frac{I}{H}$$

$$\mu_r = 1 + \chi$$

This is the required relation.

13. Retentivity or Remanence

When an external field is applied to the specimen it is magnetized and when the field is removed it is demagnetized. But some materials do not completely demagnetize when field is removed. There is some magnetism left out in the specimen. “This residual magnetism which is left out even after the removal of the external magnetic field” is called as the Retentivity or Remanence.

14. Coercivity

The residual magnetism can be removed completely from the material by applying a reverse magnetic field. “The reverse magnetic field which is used to completely remove the residual magnetism” is called as the coercivity.

3.2 ORIGIN OF MAGNETIC MOMENTS

The macroscopic magnetic properties of a substance are a consequence of magnetic moments associated with individual electrons. Each electron in an atom has magnetic moments that originate from the following two sources

1. Orbital magnetic moment of electrons
2. Spin magnetic moment of electrons.

Magnetic moments associated with an orbiting electron and a spinning electron is shown in Fig.3.3 (a and b).

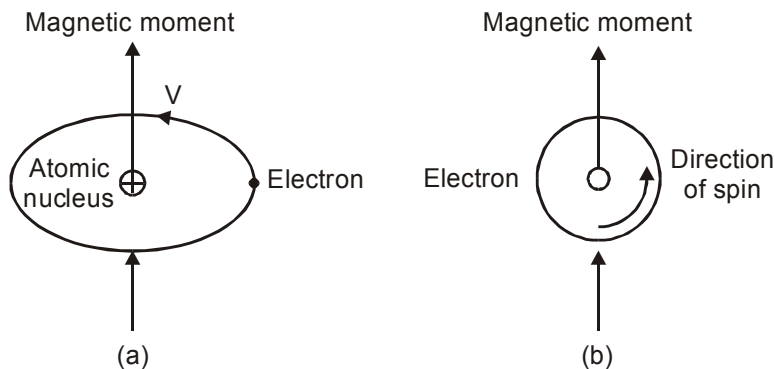


Fig. 3.3 Magnetic moments

We know that the electrons in an atom revolve around the nucleus in different orbits. Basically, there are three contributions for the magnetic dipole moment of an atom.

The orbital motion of electrons (the motion of electrons in the closed orbits around the nucleus). It is called as orbital magnetic moment. Its magnitude is always small. Spin motion of the electrons (i.e. due to electron spin angular momentum) and it is called as spin magnetic moment.

The contribution from the nuclear spin (i.e., due to nuclear spin angular momentum). Since this is nearly 10^3 times smaller than that of electron spin, it is not taken into consideration.

For all practical purposes, we assume that the magnetic moment arises due to the electron spin ignoring the orbital magnetic moments and the nuclear magnetic moments as their magnitudes are small.

We may note that permanent magnetic moments can also arise from spin magnetic moments of the nucleus. Of all the three, the spin dipole moments of electrons are important in most magnetic materials.

1. Orbital angular momentum

This corresponds to permanent magnetic dipole moments. Let us consider an electron describing a circular orbit of radius 'r' with a stationary nucleus at the centre as shown in Fig 3.3.(a). Let the electron rotate with a constant angular velocity of 'w' radians per second.

Electron revolving in any orbit may be considered as current carrying circular coil producing magnetic field perpendicular to its plane. Thus the electronic orbits are associated with a magnetic moment. The orbital magnetic moment of an electron in an atom can be expressed in terms of atomic unit of magnetic moment called Bohr Magnetron, defined as

$$\begin{aligned} \text{1 Bohr Magnetron} \quad \mu_B &= \frac{\text{electron charge} \times \text{Planck's Constant}}{2 \times \text{Electron mass}} \\ \mu_B &= \frac{e\hbar}{2m_e} \qquad \therefore \hbar = \frac{h}{2\pi} \end{aligned}$$

2. Electron spin magnetic moment

The concept of the electron having an angular momentum has been introduced in order to explain the details of atomic spectra. This angular momentum of the electron is referred to as the spin of the electron. Since the e^- has a charge, its spin produces a magnetic dipole moment.

According to quantum theory, the spin angular momentum along a given direction is

$$\frac{+h}{4\pi} \quad \text{or} \quad \frac{-h}{4\pi}$$

Hence the spin dipole moment components along an external field are

$$\begin{aligned} \frac{eh}{4\pi m} &= 1 \text{ Bohr magnetron (or)} \\ -\frac{eh}{4\pi m} &= -1 \text{ Bohr magnetron.} \end{aligned}$$

In a many electron atom, the individual spin magnetic moments are added in accordance with certain rules. Completely filled shells contribute nothing to the resultant spin moment.

3. Nuclear magnetic moment

The angular momentum associated with the nuclear spin is also measured in units of $h/2\pi$. The mass of the nucleus is larger than that of an e^- by the order of 10^3 . Hence nuclear spin magnetic moment is of the order of 10^{-3} Bohr magnetrons.

3.3 CLASSIFICATION OF MAGNETIC MATERIALS

The magnetic materials are classified into two categories:

1. The materials without permanent magnetic moment

Example: 1. Diamagnetic materials.

2. The materials with permanent magnetic moment.

Example: 1. Paramagnetic materials

2. Ferromagnetic materials

3. Anti-Ferromagnetic materials

4. Ferrimagnetic materials.

3.3.1 Diamagnetic materials

Definition

In a diamagnetic material the electron orbits are randomly oriented and the orbital magnetic moments get cancelled. Similarly, all the spin moments are paired i.e., having even number of electrons. Therefore, the electrons are spinning in two opposite directions and hence the net magnetic moment is zero.

Effect of magnetic field

When an external magnetic field is applied, the electrons re-orient and align perpendicular to the applied field, i.e., their magnetic moment opposes the external magnetic field.

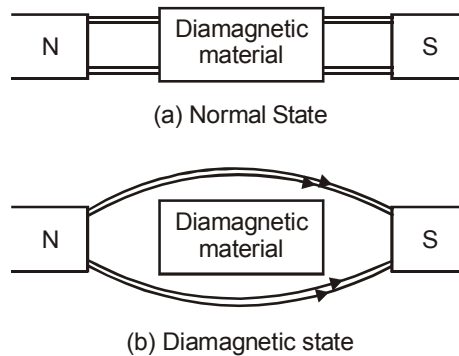


Fig. 3.4 Effect of magnetic field in Diamagnetic material

In the above diagram, there is no penetration of magnetic lines through the diamagnetic material.

Properties

1. They repel the magnetic lines of force, if placed in a magnetic field as shown in figure (3.4).
2. The susceptibility is negative and it is independent to temperature and applied field strength. ($\chi = -ve$)
3. The permeability is less than one ($\mu < 1$)
4. There is no permanent dipole moment.

5. When the temperature is greater than the critical temperature diamagnetic becomes normal material.
6. It has superconducting property.

Examples : Gold, germanium, silicon, antimony, bismuth, silver, lead, copper, hydrogen, Water and alcohol.

3.3.2 Paramagnetic Materials

Definition

Para magnetism is due to the presence of few unpaired electrons which gives rise to the spin magnetic moment. In the absence of external magnetic field, the magnetic moments (dipoles) are randomly oriented and possess very less magnetization in it.

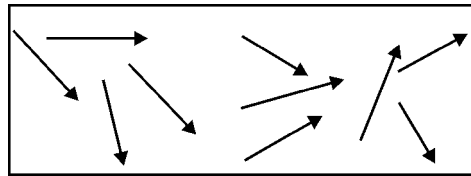


Fig.3.5 Paramagnetic Material

Effect of magnetic field

When an external magnetic field is applied to paramagnetic material, the magnetic moments align themselves along the field direction and the material is said to be magnetized. This effect is known as paramagnetism.

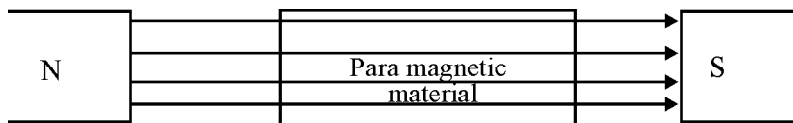


Fig. 3.6 Effect of magnetic field in paramagnetic material

Thermal agitation disturbs the alignment of the magnetic moments with an increase in temperature, the increase in thermal agitation tends to randomize the dipole direction thus leading to decrease in magnetization.

This implies that the paramagnetic susceptibility decreases with increase in temperature. It is observed that the paramagnetic susceptibility varies inversely with temperature.

$$\chi \propto \frac{1}{T} \therefore \chi = \frac{C}{T}$$

This is known as the Curie law of paramagnetism.

Properties

1. The magnetic lines of force pass through the materials.
2. The susceptibility is positive and is given by $\chi = C / (T - \theta)$ (Curie-Weiss law)
3. The permeability is greater than one ($\mu > 1$).
4. There is a permanent magnetic moment.
5. When the temperature is less than the Curie temperature, paramagnetic materials become diamagnetic materials.
6. Its spin alignment is random in nature.

Examples : Platinum, CuSO_4 , MnSO_4 , Aluminum, etc

3.3.3 Ferromagnetic materials

Definition

Ferromagnetism is due to the presence of more unpaired electrons. Even in the absence of external field, the magnetic moments align parallel to each other. So that it has large magnetism. This is called spontaneous magnetization.

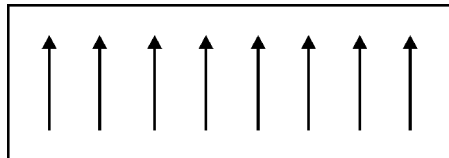


Fig. 3.7 Ferromagnetic materials

Effect of magnetic field

If a small external magnetic field is applied the magnetic moments align in the field direction and become very strong magnets.



Fig.3.8 Effect of magnetic field in ferromagnetic material

Properties of ferromagnetic materials

1. All the magnetic lines of force pass through the material.
2. Its susceptibility is **high positive** and it is given by $X = \frac{C}{T - \theta}$
3. The permeability is very much greater than one.
4. They have enormous permanent dipole moment.
5. When the temperature is greater than the Curie temperature, the Ferromagnetic material becomes paramagnetic material.
6. The ferromagnetic material has equal magnitude dipole lying parallel to each other.

Examples: Nickel, iron, Cobalt, Steel, etc.

(**Curie temperature** - The temperature below which a material can acts as ferromagnetic material and above which it can acts as paramagnetic material is called Curie temperature.)

3.3.4 Dia, Para and Ferro magnetic materials – Comparison

<i>S. N</i>	<i>Properties</i>	<i>Diamagnetic</i>	<i>Paramagnetic</i>	<i>Ferromagnetic</i>
1	Definition	It is a material in which there is no permanent magnetic moment.	It has permanent magnetic moment.	It has enormous (more) permanent magnetic moment.
2	Spin or magnetic moment or dipole alignment.	No spin alignment.	Random alignment	Parallel and orderly alignment.
3	Behavior	Repulsion of magnetic lines of force from the centre of the material.	Attraction of magnetic lines towards the centre.	Heavy attraction of lines of force towards the centre.
4	Magnetized direction	Opposite to the External magnetic field.	Same direction as the External magnetic field.	Same direction as the External magnetic field.
5	Permeability	It is very less	It is high	It is very high
6	Relativity permeability	$\mu_r < 1$	$\mu_r > 1$	$\mu_r \gg 1$
7	Susceptibility	Negative	Low positive	High positive
8	Magnetic phase transition	At 0 K, diamagnetic material is Superconductor. When we increase its temperature it becomes a normal conductor.	When temperature is less than the curie temp, it is converted in to Diamagnetic.	When temperature of the material is greater than it Curie temperature it is converted into Paramagnet.

3.4 DOMAIN THEORY OF FERROMAGNETISM

This theory was proposed by Weiss in 1907. It explains the hysteresis and the properties of ferromagnetic materials.

Magnetic Domains

A ferromagnetic material is divided into a large number of small region is called domains. (0.1 to 1 of area), each direction is spontaneously magnetized. The direction of magnetization varies from domain to domain and the net magnetization is zero, in the absence external magnetic field. The boundary line which separates two domains is called domain wall or Block wall. When the magnetic field is applied to the Ferromagnetic material, the magnetization is produced by two ways.

1. By the motion of domain walls.
2. By the rotation of domains.

Process of Domain magnetization

There are two ways to align a random domain structure by applying an external magnetic field.

1. By the motion of Domain walls

When a small amount of magnetic field is applied, the domains having dipoles parallel to the applied magnetic field increases in area by the motion of domain walls. (Fig. 3.9 (2)).

2. By the rotation of Domains

If the applied magnetic field is further increased, the domains are rotated parallel to the field direction by the rotation of domains. (fig. 3.9 (3)).

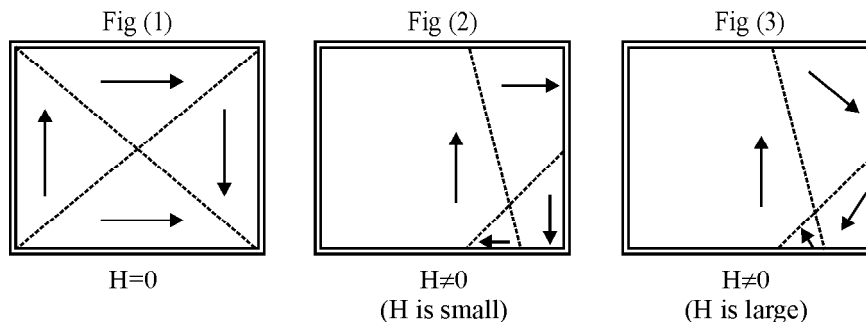


Fig. 3.9 Domin theory of ferromagnetism

3.4.1 Energies involved in the domain growth (or) Origin of Domain theory of Ferromagnetism

We can understand the origin of domains from the thermodynamic principle i.e., in equilibrium, the total energy of the system is minimum.

The total internal energy of the domain structure in a ferromagnetic material is made up from the following contributions.

1. Exchange energy (or) Magnetic field energy.
2. Crystalline energy (or) Anisotropy energy.
3. Domain wall energy (or) Bloch wall energy.
4. Magnetostriction energy

1. Exchange energy (or) Magnetic Field energy

“The interaction energy which makes the adjacent dipoles align themselves” is the called **exchange energy (or) magnetic field energy**.

The interaction energy makes the adjacent dipoles align themselves. It arises from interaction of electron spins. It depends upon the inter atomic distance. This exchange energy also called magnetic field energy is the energy required in assembling the atomic magnets into a single domain and this work done is stored as potential energy. The size of the domains for a particular domain structure may be obtained from the principle of minimum energy. The volume of the domain may vary between say, say, 10^{-2} to 10^{-6} cm^3 .

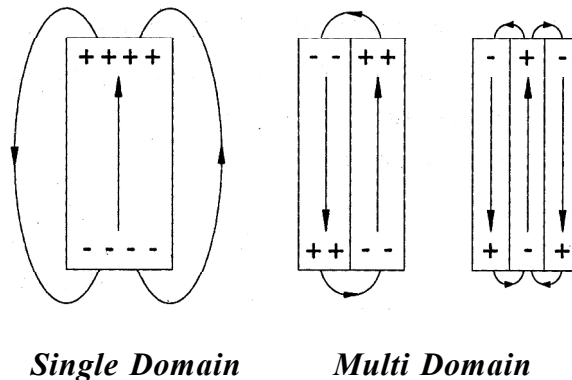


Fig. 3.10 Exchange energy in ferromagnetism

2. Anisotropy energy

The excess energy required to magnetize a specimen in particular direction over that required to magnetize it along the easy direction is called the **crystalline anisotropy energy**.

In ferromagnetic materials there are two types of directions of magnetization namely,

a) Easy direction and

b) hard directions.

In easy direction of magnetization, weak field can be applied and in hard direction of magnetization, strong field should be applied.

Crystalline anisotropy energy is energy of magnetization which is the function of crystal orientation. As shown in figure magnetization curves for iron with the applied field along different crystallographic direction crystallographic directions have been drawn. For example, in BCC iron the easy direction is $[100]$, the medium direction is $[110]$, and the hard direction $[111]$. The energy difference between hard and easy direction to magnetize the material is about. This energy is very important in determining the characteristic domain boundaries.

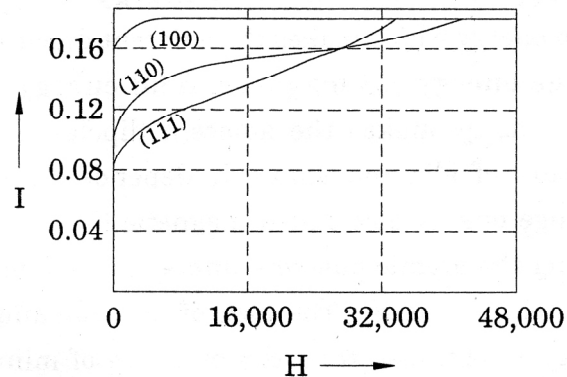


Fig. 3.11 Anisotropy energy in ferromagnetism

3. Domain wall energy or Bloch wall energy

A thin boundary or region that separates adjacent domains magnetized in different directions is called domain wall or Bloch wall.

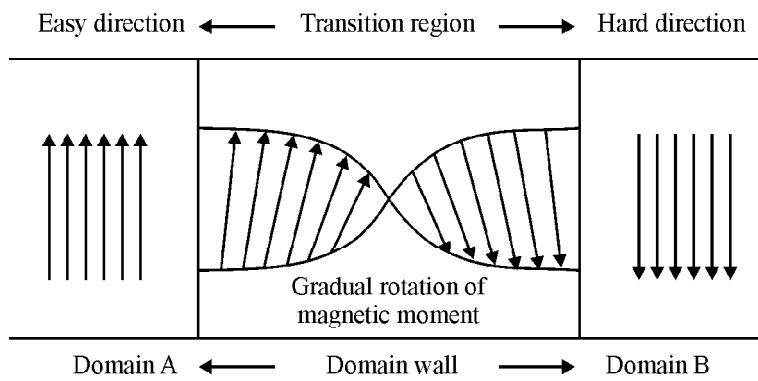


Fig. 3.12 The change of electron spin in the transition region of Bloch wall

The size of the Bloch walls is about 200 to 300 lattice constant thickness. In going from one domain to another domain, the electron spin changes gradually as shown in figure. The energy of domain wall is due to both exchange energy and anisotropic energy.

Based on the spin alignments, two types of Bloch walls may arise, namely

- a) Thick wall:** When the spins at the boundary are misaligned and if the direction of the **spin changes gradually** as shown figure, it leads to a thick Bloch wall. Here the misalignments of spins are associated with exchange energy.

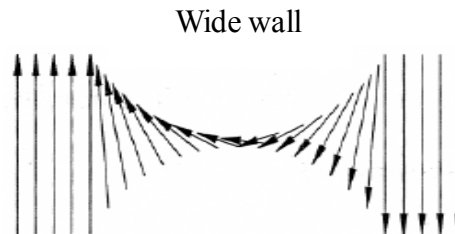


Fig. 3.13 The change of electron spin in the transition region of thick wall

- b) Thin wall:** When the spins at the boundaries **changes abruptly**, then the anisotropic energy becomes very less. Since the anisotropic energy is directly proportional to the thickness of the wall, this leads to a thin Bloch wall.

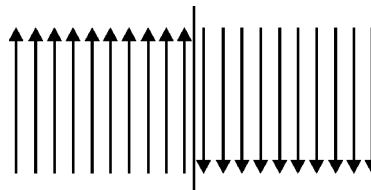


Fig.3.14 The change of electron spin in the transition region of thin wall

4. Magnetostriction energy

When a material is magnetized, it is found that it suffers a change in dimensions. This phenomenon is known as **Magnetostriction**. This deformation is different along different crystal directions. So if the domains are magnetized in different directions, they will either expand or shrink. This means that work must be done against the elastic restoring forces. The work done by the magnetic field against these elastic restoring forces is called magneto-elastic energy or Magnetostrictive energy.

3.5 ANTIFERROMAGNETIC MATERIALS

Definition

In this material, the spins are aligned in anti-parallel manner due to unfavorable exchange interaction among them, resulting in zero magnetic moment. Even when the magnetic field is increased, it has almost zero induced magnetic moment.

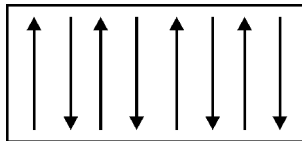


Fig. 3.15 Spin alignment of antiferromagnetic materials

Properties

1. Its susceptibility is very small and it is positive.
2. The susceptibility is given by $\chi = \frac{C}{T + \theta_N}$ for $T > T_N$ [where T_N – Neel temp]
3. Initially, when the temperature increases, susceptibility $[\chi]$ value of the anti Ferro magnetic material also increases and reaches a maximum at a particular temperature this temperature called as **Neel temperature**, susceptibility decreases with increase in temperature and the material changes into paramagnetic material. (**Neel temperature** - The temperature at which susceptibility is maximum is called Neel temperature).

Examples : Ferrous oxide, FeCl_4 , MnO_4 , MnS and some ionic compounds etc.

3.6 FERRIMAGNETIC MATERIALS

Definition

Ferrimagnetic materials or Ferrites are much similar to Ferromagnetic materials. The magnetic dipoles are aligned anti-parallel with unequal magnitudes. If small value of magnetic field is applied, it will produce the large value of magnetization.

Ferrimagnetic materials are widely used in high frequency applications and computer memories.

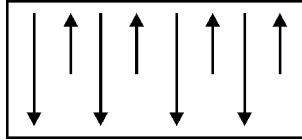


Fig. 3.16 Spin alignment of Ferrimagnetic materials

Properties

1. The susceptibility is very large and positive and it is given by, $\chi = \frac{C}{T \pm \theta}$
when $T > T_N$
2. Beyond the Neel temperature susceptibility (χ) decreases.
3. These materials have low eddy current loss and low hysteresis losses.

Examples: Ferrous Ferrites and Nickel Ferrites

3.7 HYSTERESIS

Hysteresis means “Lagging” i.e., The Lagging of intensity of magnetization (I) behind the intensity of magnetic field (H).

Experimental Determination

A graph is drawn between the **intensity of magnetization [I]** and the **intensity of magnetic field [H]**, for a cycle of magnetization. The experimental setup consists of solenoid coil through which current is passed and the material is magnetized. By varying the value of current we can get different values of Intensity of magnetization [I] due to the magnetic field (H) in the solenoid.

Here $H \propto i$, $H = n i l$

Where n – no of turns in the coil

l – length of the coil and

i – Current through the circuit.

In graph, $OB = B_r =$ Retentivity

$OC = H_c =$ Coercivity

1. When the intensity of magnetic field 'H' is increased from O to F, the value of Intensity of magnetization T if also increases from O to A, at 'A' the material reaches the saturation value of Intensity of magnetization.

Then the value of I is constant.

2. When intensity of magnetic field 'H' is decreased from G to O, the value of Intensity of magnetization 'I' also decreases from A to B, but not to zero (0). Now the material retains [stores] some amount of magnetism known as **Retentivity**, even though the intensity of magnetic field 'H' is zero. It is represented as 'OB' in the graph.
3. When intensity of magnetic field 'H' is increased in reverse direction from O to C, the value of Intensity of magnetization 'I' decreases from B to C. i.e., the value of 'I' reaches zero.

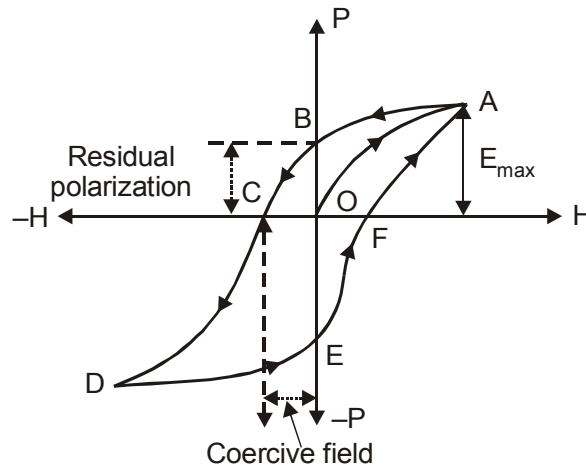


Fig.3.17 Hysteresis curve

4. The amount of intensity of magnetic field 'H' applied in the reverse direction to remove the **retentivity** is known as **Coercivity or Coercive force**. It is represented as 'OC' in the graph.
5. Further repeating the process the remaining portion [CDEFA] in the graph is obtained. The closed loop [OABCDEFA] is called Hysteresis loop (or) (I – H) curve. For one cycle of magnetization.
6. Now the material is taken out. After a cycle of magnetization, there is some expenditure (loss) of energy.
7. This loss of energy is radiated in the form of heat energy in the material.
8. This loss of energy is directly proportional to the area of the loop.
9. **From the Hysteresis graph, we can select soft and hard magnetic materials depending upon the purpose.**

Energy product

It is the product of residual magnetism B_r and coercivity which H_c gives the maximum amount of energy stored in the specimen.

$$\text{Energy product} = B_r \times H_c$$

Hysteresis loss

When the specimen is taken through a cycle of magnetization. There is a loss of energy in the form of heat. This loss of energy is known as Hysteresis loss.

3.7.1 Explanation of hysteresis on the basis of Domains

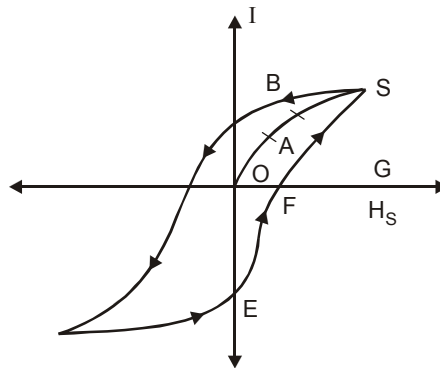


Fig.3.18 Hysteresis curve on the basis domain theory

- OA - Due to smaller reversible domains wall movement.
- AB - Due to larger irreversible domain wall movement.
- BS - Due to smaller irreversible domain rotation.
- S - Point of saturation.

1. When a field is applied, for small H, the domain walls are displaced and gives rise to small value of magnetization. [OA in the graph]. Now, the field is removed, the domains return to its original state known as reversible domains.
2. When the field is increases, a large number of domains contribute to the magnetization and I increases rapidly with H.
3. Now, when the field is removed the domain boundaries do not come back to the original position due to the domain wall movement to a very large distance (AB in the graph). These domains are called irreversible domains.
4. Now if the field is further increased, domains start rotating along the field direction and anisotropic energy is stored and it is represented as BC in the graph.
5. Thus the specimen is said to attain maximum magnetization at this position even after the removal of the field. The material is having magnetism called Retentivity. This Retentivity can be destroyed by applying a high reverse magnetic field called coercivity.
6. Thus the reversible and irreversible domain wall movements give rise to hysteresis in the Ferromagnetic materials.

3.8 HARD AND SOFT MAGENTIC MATERIAL

In Hysteresis, after a cycle of magnetization, there is some expenditure (loss) of energy. This loss of energy is radiated in the form of heat energy in the material and it is directly proportional to the area of the loop. From the Hysteresis graph, we can select the soft and hard magnetic materials.

3.8.1 Hard Magnetic Materials

The materials which are very difficult to magnetize and demagnetize are called hard magnetic materials. These materials can be made by heating the magnetic materials and then cooling it **suddenly**. It can also be prepared by adding impurities.

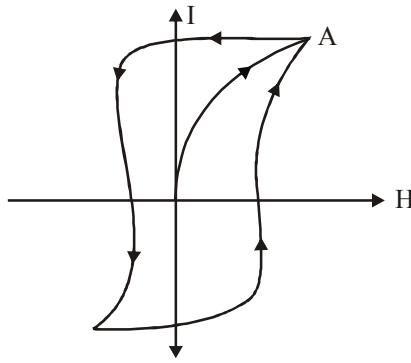


Fig.3.19 Hysteresis loop of a hard magnetic material

The above hysteresis loop is very hard and has a large loop area for a hard magnetic material, therefore the loss is also large. Domain wall does not move easily and require large value of H for magnetization. Its coercivity and retentivity values are large. Its eddy current loss is also high due to its low resistivity, the magnetostatic energy is large. It has low susceptibility and permeability. The hard magnetic materials have large amount of impurities and lattice defects.

Examples : *Tungsten steel, Carbon steel, Chromium steel, Alnico etc.,*

Properties

- 1) It is easily magnetised and demagnetised.
- 2) Their hysteresis area is very small and hence, the hysteresis loss is also small, as shown in figure.
- 3) The coercivity and retentivity are very small.
- 4) These materials have large values of susceptibility and permeability.
- 5) Their magnetostatic energy is very small.
- 6) The eddy current loss is very small.

Applications

- 1) Iron-Silicon alloys are used in electrical equipment and magnetic cores of transformers.
- 2) Cast iron is used in the structure of electrical machinery and the frame work of D.C machine.
- 3) Nickel alloys are used to manufacture inductors, relays and small motors.
- 4) It is also used for computer and data storage devices.

3.8.2 Soft Magnetic Materials

The materials which are easily magnetized and demagnetized are called **soft magnetic materials**. These materials can be made by **heating** the magnetic materials and then cooling it **slowly** to attain an ordered structure of atoms.

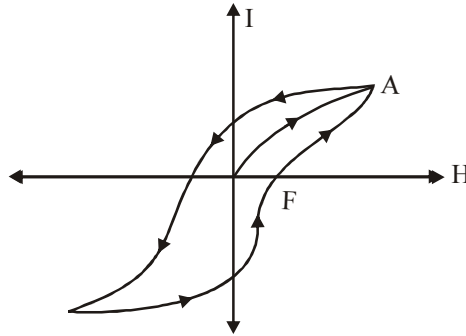


Fig.3.20 Hysteresis loop of a soft magnetic material

The above hysteresis loop is very small and has a less loop area for a soft magnetic materials. Therefore the loss is also small. Domain wall move easily and require small value of H for magnetization. Its coercivity and retentivity values are small, its eddy current loss is small due to its high resistivity. The magnetostatic energy is less, it has high value of susceptibility and permeability. The soft magnetic materials do not have impurities and lattice defects.

Examples: *Iron-Silicon alloys, Nickel-Iron alloys and Iron-cobalt alloys.*

Properties

- 1) It is very hard to magnetize and also demagnetize.
- 2) The hysteresis cure is very broad and has a large as shown in figure.
- 3) The coercivity and retentivity values are large.
- 4) These materials have low value of susceptibility and permeability.
- 5) The magnetostatic energy is large.
- 6) The eddy current loss is very high.

Applications

- 1) Magnets made by carbon steel are used for manufacturing the toys and compass needle.
- 2) Tungsten steel is used in making permanent magnets for D.C motors.
- 3) It is also used for making a small size of magnets.

3.8.3 Difference between Hard and Soft magnetic materials

<i>S. No</i>	<i>Hard Magnetic Materials</i>	<i>Soft Magnetic Materials</i>
1.	Cannot be easily magnetized	Can be easily magnetized.
2.	It can be produced by heating and sudden cooling	It can be produced by heating and slow cooling.
3.	Domain wall does not move easily and require large value of H for magnetization.	Domain wall move easily and requires small value of H for magnetization.
4.	Hysteresis loop area is large Susceptibility and Permeability values are small.	Hysteresis loop area is small Susceptibility and Permeability values are high.
5.	Retentivity and Coercivity are large	Retentivity and Coercivity are small.
6.	High eddy current loss	Low eddy current loss
7.	Impurities and defects will be more	No impurities and defects
8.	Examples: Alnico, Chromium steel, tungsten steel, carbon steel.	Examples: Iron-silicon alloy, Ferrous nickel alloy, Ferrites Garnets.
9.	Uses: Permanent magnets, DC magnets.	Uses: Electro magnets, computer data storage. Transformer core.

3.9 FERRITES

Definition

Ferrites or Ferrimagnetic materials are the modified structure of iron without carbon. In Ferrities the spins of adjacent ion is the presence of a magnetic field are in opposite directions with different magnitudes.

3.9.1 Properties

1. These are made from ceramic ferromagnetic compounds.
2. It has low tensile strength and it is brittle and soft.
3. In these materials all valence electrons are tied up by ionic bonding.
4. These are bad conductors with high resistivity of the order of $10^{11} \Omega\text{m}$
5. Ferrites have low eddy current loss and low hysteresis loss.

6. The general formula for Ferrites is $X^{2+} (Fe_2)^{3+} O_4$ where X may is a metal (divalent metal) such as Mg, Ni, Mn, Zn, etc.
7. Ferrites are manufactured by powder metallurgical process by mixing, compacting and then sintering at high temperatures followed by age hardening in magnetic fields.

3.9.2 Structures of Ferrites

Ferrites are the magnetic compounds consisting to two or more different kinds of atoms. Generally ferrites are expressed as $X^{2+} (Fe_2)^{3+} O_4$ where X^{2+} stands for suitable divalent metals ions such etc Mg^{2+} , Zn^{2+} , Fe^{2+} , Mn^{2+} , Ni^{2+} etc.

Normally, there are two types of structures present in the ferrites

1. Regular spinel
2. Inverse spinel

3.9.3 Regular spinal

In the regular spinal type, each metal atom (divalent) is surrounded by four ions in a tetragonal fashion.

For example in Mg^{2+} , Fe_2^{3+} , O_4^{2+} , the structure of Mg^{2+} is given in the Fig. 3.21 and it is called “A’ site. Totally in a unit cell, there will be 8 tetrahedral (8A) sites.

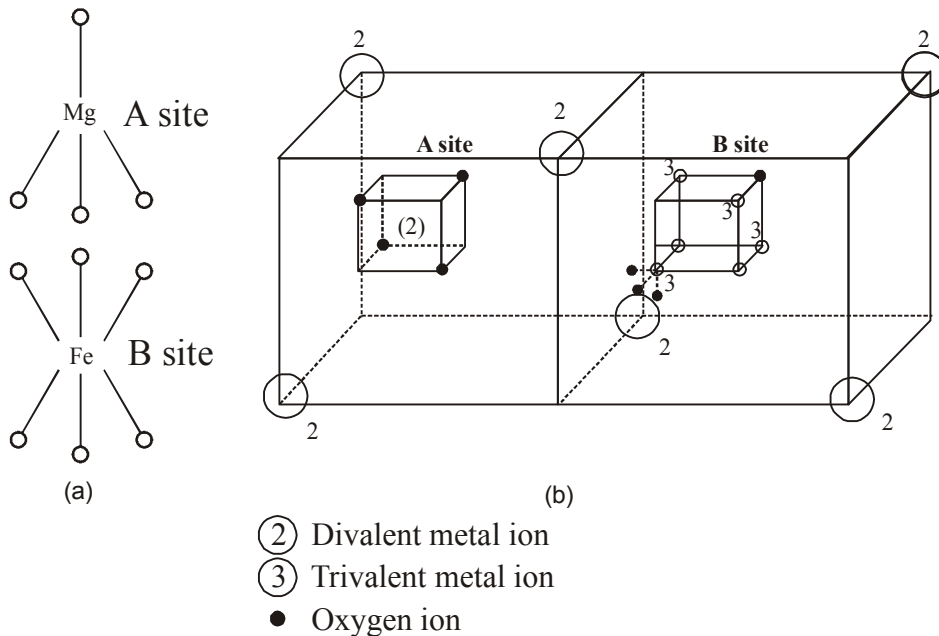


Fig 3.21 Structure of ferrites (Regular Spinal)

Each Fe_{3+} (trivalent) is surrounded by '6' O^{2-} ions and forms an octahedral fashion as show in Fig.3.20. Totally there will be 16 such octahedral sites in the unit cell. This in indicated by 'B' site.

Thus in a regular spinal, each divalent metal ion (Mg^{2+}) exists in a tetrahedral form (A site) and each trivalent metal ion (Fe^{2+}) exists in an octahedral form (B site). Hence, the sites A and B combine together to form a regular spinal ferrite structure as shown in Fig.3.21.

3.9.4 Inverse spinal

In this type, we consider the arrangement of dipoles of a single ferrous ferrite molecule $\text{Fe}^{3+} [\text{Fe}^{2+} \text{Fe}^{3+}] \text{O}_4 \text{Fe}^{3+}$, ions (trivalent) occupies all A sites (tetrahedral) and half of the B sites (octahedral) also.

Thus the left out B sites will be occupied by the divalent (Fe^{2+}). The inverse spinal structure is shown in the Fig. 3.22.

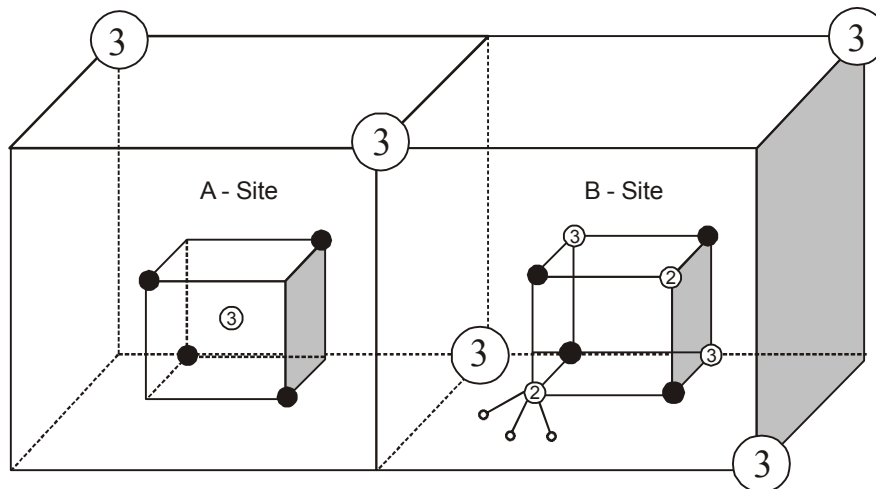


Fig.3.22 Structure of ferrites (inverse Spinal)

3.9.5 Types of interaction present in the ferrites

The spin arrangement between the A site B site is in an ant parallel manner and it was explained by Neel. According to him, in ferrites, the spin arrangement is ant parallel and there exists some interaction between the A and B sites which is represented as AB interaction.

Apart from this, there are two more interactions (i.e.,) AA and BB interaction which is negative and considerable weaker than AB interaction.

The tendency of AB interaction is to align all A spins parallel to each other and anti parallel to all B spins, but the tendency of AA and BB interaction is to spoil the parallel arrangement of AB spins respectively.

Since AB is very strong as compared with AA and BB, the effect of AB interaction dominates and gives rise to anti parallel spin arrangement.

3.9.6 Properties of ferromagnetic materials

1. Ferromagnetic materials possess net magnetic moment.
2. Above Curie temperature, it becomes paramagnetic while it behaves as ferromagnetic material below Curie temperature.
3. The susceptibility of Ferrimagnetic material is very large and positive. It is temperature dependent and is given by

$$\chi = \frac{C}{T \pm \theta} \quad \text{for } T > T_N$$

4. Beyond Neel temperature, decreases.
5. Spin alignment is anti parallel of different magnitudes.
6. Mechanically, it has pure ion character.
7. They have high permeability and high resistivity.
8. They have low eddy current losses and low hysteresis.

3.9.7 Application of Ferrites

1. Ferrites are used to produce ultrasonic wave by Magnetostriction principle.
2. Ferrites are used in audio and video transformers.
3. Ferrites rods are used in radio receivers to increase the sensitivity.
4. Ferrites are widely used in non-reciprocal microwave devices such as gyrator, circulator and Isolator.

Gyrator : It transmits the power freely in both directions with a phase shift of radians.

Circulator : It provides sequential transmission of power between the ports.

Isolator : It is used to display differential attenuation.

5. They are also used for power limiting and harmonic generation.
6. Ferrites are used in parametric amplifiers so that the input can be amplified with low noise.
7. They are used in computers and data processing circuits.
8. Bi-stable elements, Ferro cube (Ferrite with square hysteresis loop), magnetic shift register, and magnetic bubbles are also examples for Ferrites.

3.10 MAGNETIC RECORDING AND READOUT MEMORY

Nowadays, large number of information are stored in (or) retrieved from the storage devices, by using devices is magnetic recording heads and they function according to the principle of magnetic induction.

Generally Ferro or Ferrimagnetic materials are used in the storage devices because in this type of materials only the magnetic interaction between only two dipoles align themselves parallel to each other.

Due to this parallel alignment even if we apply small amount of magnetic field, a large value of magnetization is produced. By using this property information are stored in storage devices.

In the storage devices, the recording of digital data (0's and 1's) depends upon the direction of magnetization in the medium.

3.10.1 Magnetic parameters for Recording

1. When current is passed through a coil, a magnetic field is induced. This principle called "Electromagnetic Induction" is used in storage devices.
2. The ease with which the material can be magnetized is another parameter.
3. We know the soft magnetic materials are the materials which can easily be magnetized and demagnetized. Hence a data can be stored and erased easily. Such magnetic materials are used in temporary storage devices.
4. Similarly, we know hard magnetic materials cannot be easily magnetized and demagnetized. So such magnetic materials are used in permanent storage devices.
5. In soft magnetic materials, the electrical resistance varies with respect to the magnetization and this effect is called magneto-resistance. This parameter is used in specific thin film systems.

The magnetic medium is made of magnetic materials (Ferro or Ferric oxide) deposited on this plastic.

The magnetic medium move across the read / write heads and either logic 1's and logic 0's are written on the medium. The magnetized spots on the medium generate small electrical signals and this different direction signals represents logic 1's and 0's on the medium.

3.10.2 Storage of Magnetic Data

Memory units are the devices used to store the information (Input and Output) in the form of bits [8bits = 1 Byte].

The memory units are classified into two categories.

1. Main memory (Primary) or Internal Memory.
2. Auxiliary Memory (Secondary) or External Memory.

1. Main Memory

The memory unit of the central processing unit (CPU) is called as main memory. Compare a black beard main memory. We can write many data on memories and finally erase it if we want.

Example : RAM, ROM, EPROM etc.

2. Auxiliary Memory

Since storage capacity of primary memory is not sufficient secondary memory units are developed to store the large volume of data. Separately and hence called as extra (or) additional (or) external memory.

This type of memory is also referred to as back up storages because, it is used to store large volume of data on a permanent basis.

Example:

1. Magnetic tapes
2. Magnetic disk (Floppy and Hard disc)
3. Ferrite core memories
4. Magnetic bubble memories.

3.10.3 Magnetic Tape

It is one of the most popular storage medium for data. The tape is a plastic ribbon with metal oxide material coated on one side which can be magnetized. In this, information can be written and also can be read by read / write heads.

Information recorded in the tape is in the form of tiny magnetized and non magnetized spots on the metal oxide coating.

The magnetized spot represents '1' and un magnetized spot represent '0' in binary code. The information can be accessed, processed, erased and can be again stored in the same area.

Advantages

1. Its storage capacity is large
2. It is easy to handle and is portable
3. Its cost is less than other storage devices.
4. It can be erased and reused many times.

Disadvantage

1. It consumes lot of time.

3.10.4 Magnetic Disc Drivers

These disks are direct access storage devices. These disks are magnetically coated. There are two types of disks.

1. Hard disc
2. Floppy disc

1. Hard disc

The hard disc is made of hard aluminum platters. The platter surface is carefully machined until it is flat (or) plane. The platter surface is coated with magnetic material (magnetic oxides). The platter is built into a box.

Similar such disks are mounted on a vertical shaft, forming a disk pack and it is shown in fig.

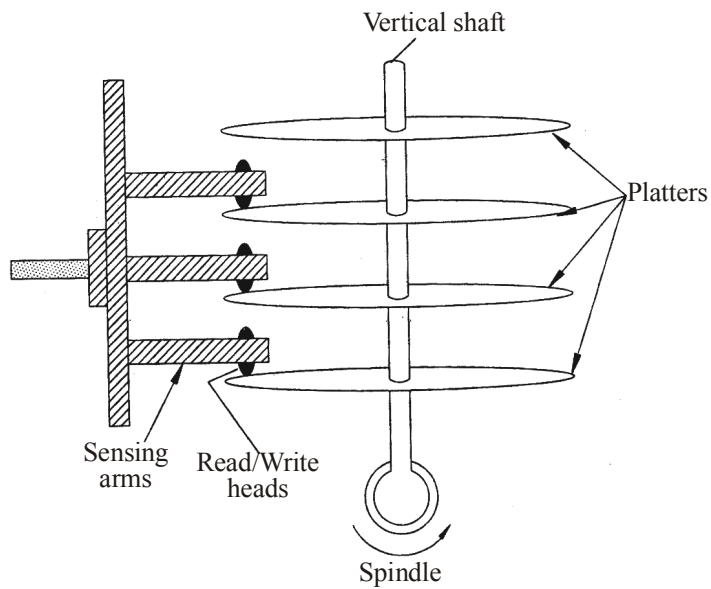


Fig.3.23 Hard disc

The disc pack is placed in a drive mechanism called hard disk drive. The drive mechanism drives the disc pack with the spindle. The data is written (or) read by R/W heads in the horizontal sensing arms by moving in and out between the platters with the precaution that the R/W head doesn't touch the surface; instead, it flies over the disk surface by a fraction of an mm.

Advantages

1. It has very large storage capacity.
2. Thousands of files can be permanently stored.
3. Very high speed in reading and writing the information.
4. This is prevented from dust particles, since they are sealed in a special chamber.

Disadvantages

1. It is very costly.
2. If data is once corrupted, there is a heavy loss of data.

3.10.5 Floppy Disk

The hard disc is suitable only for large and medium sized computers and often are too expensive for small computers systems. Floppy disc are the latest development in secondary storage devices.

It is made up of a flexible plastic material and hence called as floppy disc. It is also called as diskette. It acts both as an input and output device.

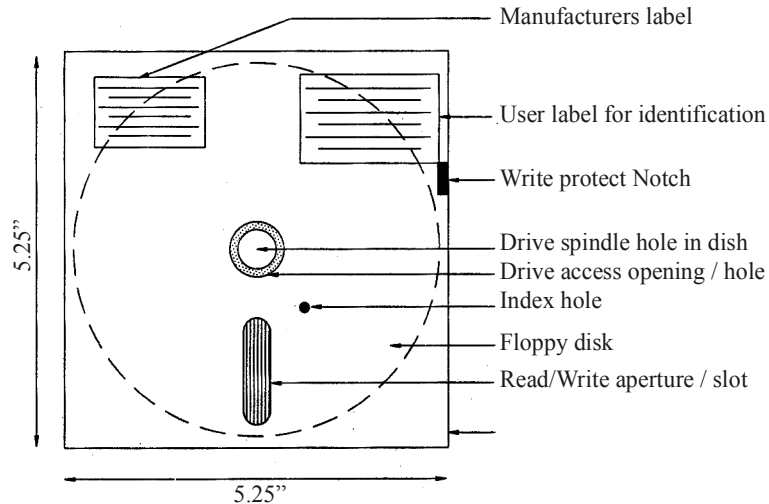


Fig.3.24 floppy disc

The disk is provided with a central hole. This hole is used for mounting the disc in the floppy drive unit. The envelope prevents the disk from dust and moisture.

There is a small index hole in the cover and there will be a hole in the drive disk. When these two holes match then only the storage operation can be started. Write protect notch is used to prevent writing on the disc by other users. This can be done by covering the notch with a sticker. A 5.25" floppy is shown in fig.

Writing operation

When the floppy disk moves over the gap the CPU flow through the write will of the head and magnetizes the iron oxide coating in the disk to the proper pattern.

Reading Operation

When the data are to be read, the magnetized patterns induces pulses of current in the read coil and is amplified then fed to the CPU. Thus data can be stored and accessed from the floppy disc on both sides (or) single side. Reading / writing data on the magnetic medium using frequently modulated wave.

Special features

1. The cost is very low
2. It can be easily handled
3. It can be taken to any place
4. It has high storage capacity
5. Many types of floppies are available, depending on their storage capacities.

Disadvantage

Here the magnetic disk is moved (rotated) mechanically.

3.10.6 Magnetic bubble Materials

Magnetic bubble is direct access storage medium, magnetic bubbles are soft magnetic materials with magnetic domains of a few micrometers in diameter. These bubbles are the electrical analogue of the magnetic disk memories used in computers.

The magnetic disk in the hard disk memory is moved mechanically where as the bubbles in a bubble memory device are moved electronically at very high speeds, so the read out time or storing time is greatly reduced in bubble memory device. The bubble units are made with solid state electronic chips.

A magnetic bubble can be thought of as a positively charged and in a negatively charged magnetic film. The presence of a bubble is on 'ON' condition and the absence of a bubble is an 'off' condition. ie., [1 or 0].

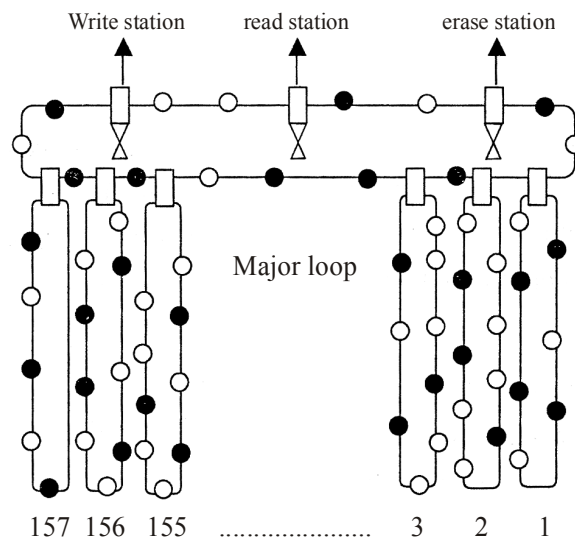
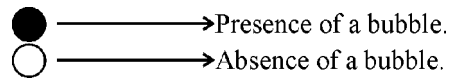


Fig. 3.25 Magnetic bubble memory

Figure shows the schematic diagram of a magnetic bubble memory. It consists of one major loop and 157 minor loops. Each minor loop has 641 bubble sites.



Writing operation

When a data is to be stored, the bubbles from the minor loops are transferred to the major loop, and it goes to the write station. In write station the data is entered and the bubble again comes to minor loop.

Reading operation

To read the data from the storage, the bubble from minor loops are transferred to the major loop and it goes to the read station, then it comes to the minor loop. The data can be altered by the erase station, if we need to erase it.

Advantage

1. It is non-volatile.
2. It has high storage capacity than the magnetic hard disk.
3. It has high across speed.

Construction

A Bubble memory consist of materials such as magnetic garnets and store the data as microscopic magnets. A thin film of these garnets is deposited on a non-magnetic substrate of Gadolinium Gallium Garnet in Integrated Circuit [IC] form.

SUPER CONDUCTORS

3.11 INTRODUCTION TO SUPERCONDUCTIVITY

It was thought that the electrical resistance of a conductor becomes zero only at absolute zero temperature. But in 1911, H. Kammerlingh Onnes studied the properties of mercury at very low temperature using liquid helium and is found that the resistivity of mercury drops to zero at 4.2 K and changes into a superconducting material.

Mathiessen's rule: It states that “the electrical resistivity $[\rho]$ of the material is the sum of the residual resistivity due to scattering by impurities $[\rho_0]$ and the ideal resistivity due to scattering by phonons $[\rho_T]$.”

$$\text{ie., } \rho = \rho_0 + \rho_T$$

Definition

The ability of certain metals and alloys exhibit almost zero electrical resistivity when they are cooled to low temperature is known as superconducting. (ie., maximum conductivity with zero resistance at zero Kelvin)

Each of these parameters is very dependent on the other two properties present.

Critical temperature (T_C) (or) Transition Temperature

The temperature at which a normal conductor loses its resistivity and becomes a superconductor is known as critical temperature (or) Transition temperature. Every superconductor has its own critical temperature at which it passes over into superconducting state. Depending on the transition temperature, superconductors are classified into two groups are

i) Low temperature superconductors (LTS): The superconductors which have low transition temperature (below 30K) are known as low temperature superconductors.

Example: Tin (3.2 K), Mercury (4.15 K).

ii) High temperature superconductors (HTS): The superconductors which have high transition temperature (above 30K) is known as high temperature superconductors.

Example: Barium - Lanthanum - Copper - Oxide (BLCU) - 35 K

Yttrium - Barium - Copper - Oxide - ($\text{YBa}_2\text{Cu}_3\text{O}_4$) - 92 K

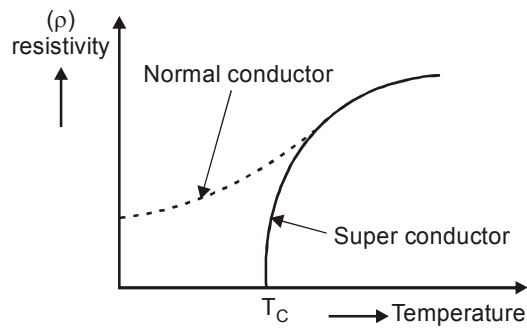


Fig. 3.26 Variation of electrical resistivity with temperature

3.12 PROPERTIES OF SUPERCONDUCTORS

At Critical temperature, the following properties are observed.

1. The electrical resistivity drops to zero.
2. The magnetic flux lines are excluded (ejected out) from the superconductors.
3. There is discontinuous change in the specific heat.
4. There are small changes in the thermal conductivity and volume of the materials.

3.12.1 Critical magnetic field (Magnetic Property)

A very strong magnetic field applied to superconducting material it disappears super conducting property this is called as critical magnetic field.

$$H_C = H_0 \left[1 - \left[\frac{T}{T_C} \right]^2 \right] \quad H_0 \text{ Where } = \text{Critical field at } 0 \text{ K}$$

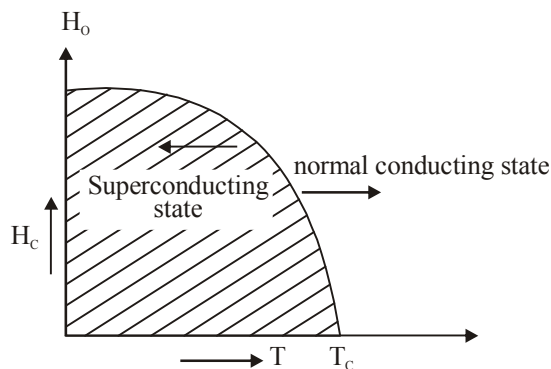


Fig.3.27 Critical magnetic field

It is noted that when the temperature of a material increases, the value of critical magnetic field decreases. Therefore the value of critical magnetic fields are different for different materials.

3.12.2 Diamagnetic property (Meissner effect)

If a normal conducting material is placed in a magnetic field of flux density B , the magnetic lines penetrate through the material.

Now the material is cooled below its transition temperature when $T \leq T_c$ then the magnetic lines of forces are expelled out from the material as shown in figure.

We know that, a diamagnetic material have the tendency to expel the magnetic lines of force. Since the superconductor also expels the magnetic lines of force and it behaves as a perfect diamagnet. This behaviour was first observed by Meissner and hence called as Meissner effect.

i) Meissner effect.

When the superconducting material is subjected to a uniform magnetic field, under the condition $T \leq T_c$ and $H \leq H_c$, the magnetic flux lines are excluded from the material. Thus the material exhibits perfect diamagnetism. This phenomenon is called as Meissner effect.

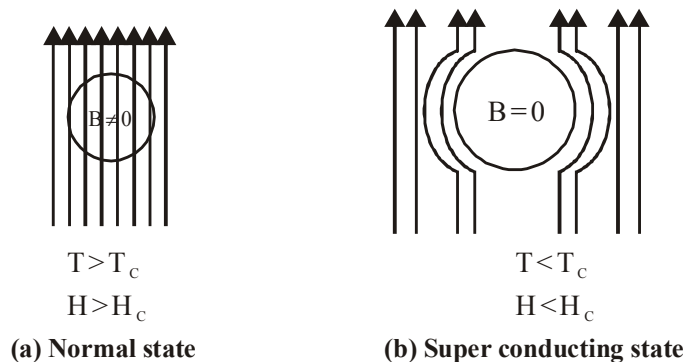


Fig. 3.28 Meissner effect

3.12.3 SQUID (Superconducting Quantum Interference Device)

(We know that a small change in magnetic field produces variation in the flux quantum.)

It consists of a superconducting ring which can have the magnetic field of quantum values (1,2,3,...) of flux placed in between two Josephson junctions as shown in figure.

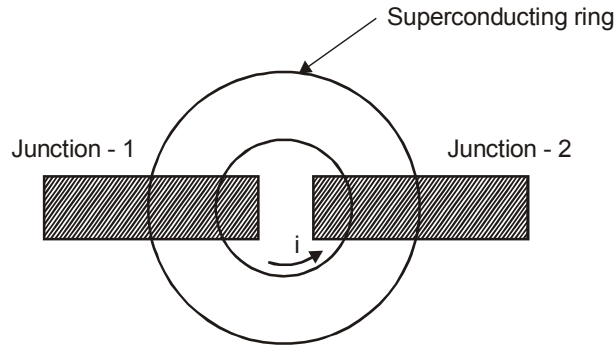


Fig. 3.29 SQUID

When the magnetic field is applied perpendicular to the plane of the ring, the current is induced at the two Josephson junctions.

The induced current produces the interference pattern and if flows around the ring so that the magnetic flux in the ring can have the quantum value of magnetic field applied.

Applications

- i) SQUID can be used to detect the variation of very minute magnetic signals in terms of quantum flux.
- ii) It is used as a storage device for magnetic flux.
- iii) It is used to study earthquakes and to remove paramagnetic impurities.

Application of Meissner effect

- 1. It is a standard test to prove whether the material is a perfect superconductor or not.
- 2. This effect is used for magnetic levitated train.

3.12.4 Effect of heavy Current

The superconducting property disappears when a heavy current flows, since current flow will set up a magnetic field.

According to Silsbee's rule, for a superconducting wire, the induced current to destroy the superconducting property is given by,

$$I_C = 2\pi r H_C$$

Where r is the radius of the superconductor.

H_C is the critical magnetic field.

3.12.5 Persistence of Current

Persistence current is one of the most important properties of a superconductor. When a current of large magnitude is induced in a superconducting ring, the current persisted in the ring even after the removal of the field at the temperature below the critical temperature, such a current flows without reducing its strength is known as persistent current.

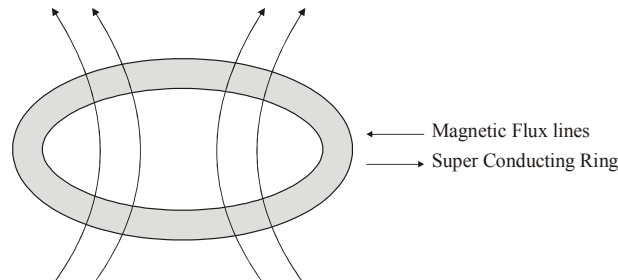


Fig. 3.30 Persistent Current

The superconducting coil with persistent current acts as a magnet. It does not require power supply to maintain its magnetic field.

3.12.6 Effect of pressure

If pressure increases, the critical temperature also increases. Therefore certain materials are brought into the superconducting state by increasing the pressure. Research is going on to get the superconducting state at room temperature by applying heavy pressure.

3.12. 7 Isotope effect

The presence of isotopes in superconductor change the transition temperature of the superconductor. The transition temperature is found to be inversely proportional to the square root of the atomic weight of the isotope (M).

$$T_c \propto \frac{1}{\sqrt{M}}$$

$$T_c = \frac{\text{const} \tan t}{\sqrt{M}}$$

$$\sqrt{M} \times T_c = \text{Constant}$$

The transition temperature of the heavier isotope is less than that of the lighter isotope.

3.12.8 General properties

1. There is no change in elastic properties, photo electric properties and crystal structure.
2. The transition temperature is unchanged with the frequency variation.

3.13 TYPES OF SUPER CONDUCTORS

Superconductors are classified as follows

Based on the value of H_c we have,

1. Type I (or) Soft superconductors
2. Type II (or) Hard superconductors

Based on the value of T_c we have,

1. High temperature superconductors
2. Low temperature superconductors

Type I Superconductor

In type I superconductor, the magnetic field is completely excluded from the material below the critical magnetic field and the material loses its superconducting property abruptly at.

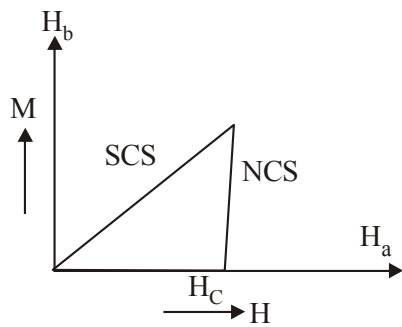


Fig.3.31 Type I Superconductor

Characteristics

1. They exhibit complete Meissner Effect.
2. They have only one critical magnetic field value.
3. Below the material behaves as superconductor and above the material behaves as normal conductor.
4. These are called as Soft superconductors.

Type II Superconductor

In type II superconductor, the magnetic field is excluded from the material and the material loses its superconducting property gradually rather than abruptly.

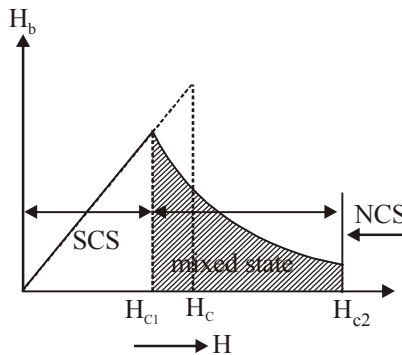


Fig. 3.32 Type II Superconductor

Characteristics

1. They do not exhibit a complete Meissner Effect.
2. They have two critical magnetic field values. Lower critical magnetic field $[H_{c1}]$ and Higher critical magnetic field $[H_{c2}]$.
3. Below H_{c1} the material behaves as superconductor and above the material behaves as normal conductor. The region in between $[H_{c1}]$ and $[H_{c2}]$ is called mixed state or vortex region.
4. These are called as Hard superconductors.

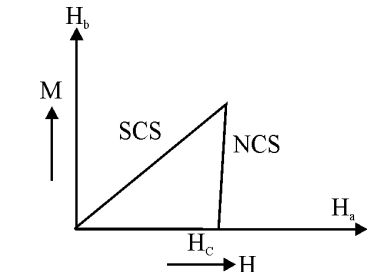
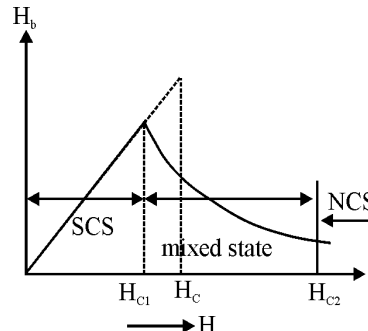
Low T_c Superconductors

1. The superconductors having the critical temperature less than 20 K are known as low T_c Superconductors or elemental superconductors.
2. The Superconductors by BCS theory.
3. It is explained by BCS theory.
4. It is not so useful due to its low temperature maintenance.
5. It is called as N-type superconductor.

High T_c Superconductors

1. The superconductors having the critical temperature greater than 100 K are known as high T_c Superconductors or ceramic or oxide superconductors.
2. The Superconductors is due to hole states.
3. It is explained by RVB theory proposed by Anderson.
4. It is very useful for commercial and engineering purposes.
5. It is called as P-type superconductor.

3.13.1 Difference between Type I and II superconductors.

S.N	Type I (or) Soft superconductors	Type II (or) Hard superconductors
1.	It exhibits complete Meissner Effect.	It does not exhibit a complete Meissner Effect.
2.	They are completely diamagnetic.	They are not completely diamagnetic.
3.	Ex : Tin, Lead, Mercury, etc.,	Ex : Nb-Zr, Nb- Ti, Nb-Sn, Va- Ga, etc.,
4.	 $H_a \rightarrow$ Applied magnetic field $H_b \rightarrow$ Induced magnetic field	
5.	The value of H_c is always too low. It is about 0.1 tesla.	The values of H_c is high and it is about 30 tesla.
6.	They have only one critical magnetic field value.	They have two critical magnetic field values. Lower critical magnetic field [H_{c1}] and Higher critical magnetic field [H_{c2}].
7.	No mixed state exists.	Mixed state is present.
8.	These are called Soft superconductors.	These are called Hard superconductors.
9.	It requires low magnetic field to destroy the superconductivity.	It requires large magnetic field to destroy the superconductivity.
10.	It loses its magnetization abruptly.	It loses its magnetization gradually.

3.13.2 Difference between High T_c and Low T_c superconductors

<i>S. No</i>	<i>High T_c Superconductors</i>	<i>Low T_c Superconductors</i>
1.	It has high T_c (>100 K).	It has low T_c (<20 K)
2.	Super conduction is due to hole states.	Super conduction is due to cooper pairs.
3.	Explained by RVB theory.	Explained by BCS theory.
4.	Very useful for commercial and engineering purposes.	It is not so useful due to its low maintenance temperature.
5.	It is called as P-type superconductor.	It is called as N-type superconductor.

3.14 HIGH TEMPERATURE (HIGH- T_c) SUPERCONDUCTORS

Superconductors with high values of critical temperature are called high temperature superconductors (HTSC). The discovery of copper oxide based ceramic materials by Bednorz and Muller in 1986 having critical temperature greater than 30 K made a new era in the field of superconductivity.

The superconductors have the critical temperature greater than 100 K are known as high Superconductors or ceramic or oxide superconductors.

The Superconductors is due to hole states and it is explained by RVB theory proposed by Anderson. It is very useful for commercial and engineering applications and it is called as P-type superconductors.

Characteristics

1. High T_c Superconductors have high temperatures.
2. They have a modified perovskite crystal structure.
3. Superconducting state is direction dependent.
4. These are oxides of copper with other elements.
5. These are reactive, brittle, and cannot be easily modified or joined.
6. For high T_c superconductors, liquid Nitrogen is used instead of liquid helium.

3.15 BCS THEORY OF SUPERCONDUCTIVITY

The properties of Type I superconductors were modeled successfully by the efforts of **John Bardeen, Leon Cooper, and Robert Schrieffer** in what is commonly called the BCS theory. A key conceptual element in this theory is the pairing of electron close to the **Fermi level** into **Cooper pairs** through interaction with the crystal lattice. This pairing results form a slight attraction between the electrons related to lattice vibrations, the coupling to the lattice is called a phonon interaction.

Pairs of electrons can behave very differently from single electrons which are fermions and must obey the Pauli Exclusion Principle. The pairs of electrons act more like bosons which can condense into the same energy level.

The electron pairs have a slightly lower energy and leave an energy gap above them on the order of 0.001 eV which inhibits the kind of collision interactions which lead to ordinary resistivity. For temperatures such that the thermal energy is less than the band gap, the material exhibits zero resistivity.

Bardeen, Cooper, and Schrieffer received the Nobel Prize in 1972 for the development of the theory of superconductivity.

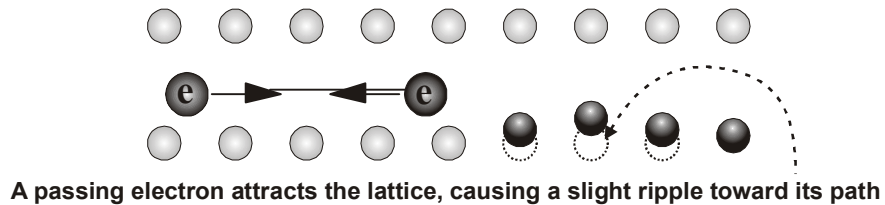
Cooper Pairs

The transition of a metal from the normal to the superconducting state has the nature of a condensation of the electrons into a state which leaves a band gap above them, this kind of condensation is seen with superfluid helium, but helium is made up of bosons-multiple electrons can't collect into a single state because of the **Pauli Exclusion Principle**.

Froehlich was first to suggest that the electrons act as pairs coupled by lattice vibrations in the material. This coupling is viewed as an exchange of phonons, phonons being the quanta of lattice vibration energy. Experimental corroboration of an interaction with the lattice was provided by the isotopic effect on the superconducting transition temperature.

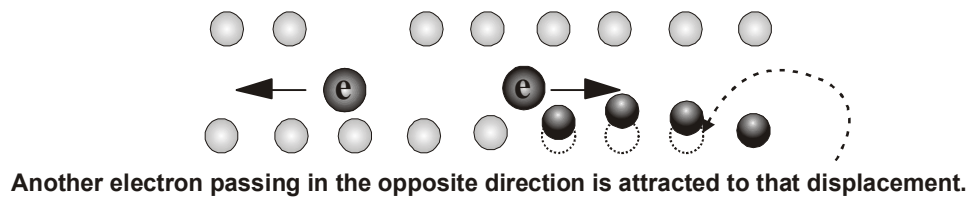
The boson-like of such electron pairs was further investigated by Cooper and they are called “**Cooper pairs**”. The condensation of Cooper pairs is the foundation of the BCS theory of superconductivity.

A model of Cooper pair attraction



Lattice of superconducting material

Fig.3.33 Cooper pair attraction



Lattice of superconducting material

Fig.3.34 Cooper pair repulsion

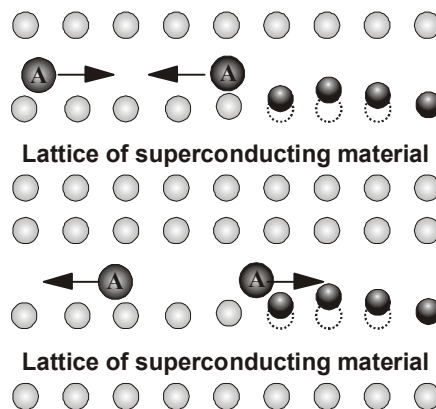


Fig. 3.35 Model of pair attraction

Ideas Leading to the BCS Theory

The BCS theory of superconductivity has successfully described the measured properties of Type I superconductors. It envisions resistance-free conduction of coupled pairs of electrons called Cooper pairs. This theory is remarkable enough that it is interesting to look at the chain of ideas which led to it.

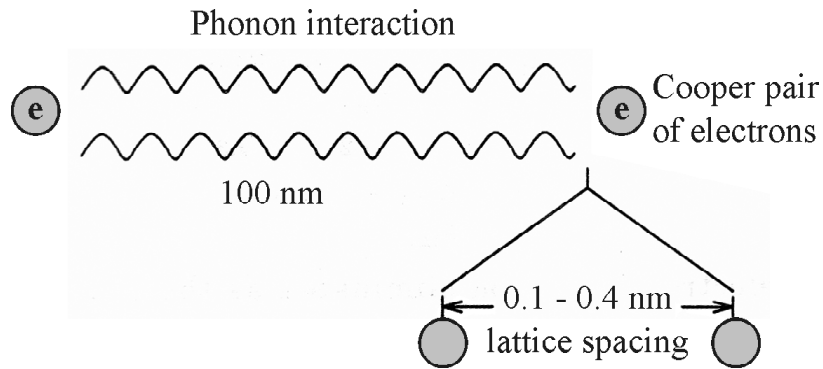


Fig. 3.36 Cooper pair (interaction)

1. One of the first steps toward a theory of superconductivity was the realization that there must be a band gap separating the charge carriers from the state of normal conduction.
 - a) A band gap was implied by the very fact that the resistance is precisely zero. If charge carriers can move through a crystal lattice without interacting at all, it must be because their energies are quantized such that they do not have any available energy levels within reach of the energies of interaction with the lattice.
 - b) A band gap is suggested by specific heats of materials like vanadium. The fact that there is an exponentially increasing specific heat as the temperature approaches the critical temperature from below implies that thermal energy is being used to bridge some kind of gap in energy. As the temperature increases, there is an exponential increase in the number of particles which would have enough energy to cross the gap.
2. The critical temperature for superconductivity must be a measure of the band gap, since the material could lose superconductivity if thermal energy could get charge carriers across the gap.

3. The critical temperature was found to depend up on isotopic mass. It certainly would not if the conduction was by free electrons alone. The made it evident that the superconducting transition involved some kind of interaction with the crystal lattice.

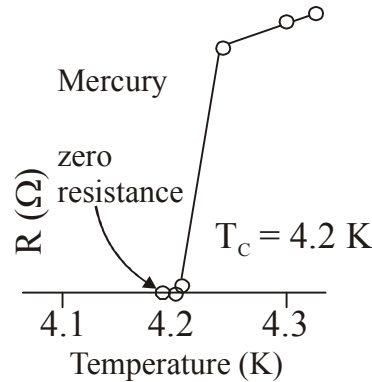


Fig. 3.37

4. Single electrons could be eliminated as the charge carriers in superconductivity since with a system of fermions you don't get energy gaps. All available levels up to the Fermi energy fill up.
5. The needed boson behavior was consistent with having coupled pairs of electrons with opposite spins. The isotope effect described above suggested that the coupling mechanism involved the crystal lattice, so this gave rise to the phonon model of coupling envisioned with Cooper pairs.

3.16 APPLICATIONS OF SUPERCONDUCTORS

General Applications

1. Electric generators can be made by using superconductors with smaller size, less weight and low energy consumption.
2. Superconductors can be used for the transmission of power over very long distances.
3. Superconductors can be used in switching Devices.
4. The superconductors can be used in sensitive electrical instruments.
5. It can be used as a memory or storage element in computers.

6. These are used to design Cryotron, Maglev, Josephson Devices and SQUID.
7. DC superconducting motors are used in ship propulsion and in large mills.
8. Superconducting magnetic field may be used to launch satellite into orbit directly from the earth without use of rockets.
9. Ore separation can be done by using machines made of superconducting magnets.
10. These are used in NMR (Nuclear Magnetic Resonance) imaging equipments which is used for scanning purposes.
11. Superconductors are used for the detection of brain tumor, defective cells, etc.,
12. Superconducting solenoids are used in magneto hydrodynamic power generation to maintain the plasma in the body.

3.17 ENGINEERING APPLICATIONS

3.17.1 Cryotron

It is a magnetically operated current switch. The superconducting property disappear when the magnetic field is greater than critical field ().

It consists of a superconducting material [A] and it is surrounded by a superconducting coil of wire [B].

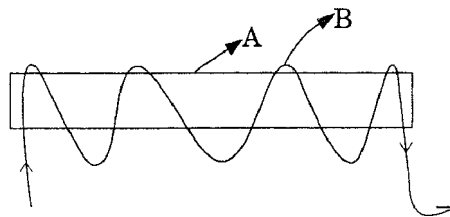


Fig. 3.38 Cryotron

When the critical magnetic field of wire B exceeds or less than that of a Superconducting material A, the current in A can be controlled by the current in the material B, it can act as relay or switching elements and it can be used as memory or storage element in computers.

3.17.2 MAGLEV (MAGnetic LEVitation)

Maglev is a magnetic levitated train, it works under the principal of Electromagnetic induction. This train cannot move over the rail. Instead it floats above the rails, so that it moves faster with speed of 500 Km/hr without any frictional loss. It has two superconducting magnet on each side of the train and there is guiding system consisting of 'S' shaped coils on each side. Due to actions of these magnets the train moves faster by levitation principle.

Working

This train consists of superconducting magnets placed on each side of the train. The train can run in a guiding system, which consists of serial 'S' shaped coil as shown in figure.

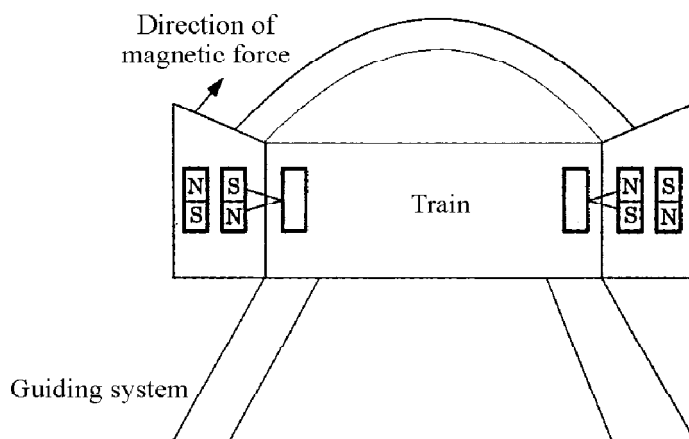


Fig. 3.39 MGLEV

Initially when the train starts, they slide on the rails. Now, when the train moves faster, the superconducting magnets on each side of the train will induce a current in the 'S' shaped coils kept in the guiding system.

This induced current generates a magnetic force in the coils in such a way that the lower half of 'S' shaped coil has the same magnetic pole as that of the superconducting magnet in the train, while the upper half has the opposite magnetic pole. Therefore, the total upward magnetic force acts on the train and the train is levitated or raised above the rails and floats in the air.

Now, by alternatively changing the poles of the superconducting magnet in the train, alternating currents can be induced in 'S' shaped coils.

Thus, alternating series of north and south magnetic poles are produced in the coils, which pulls and pushes the superconducting magnets in the train and hence the train is further moved. This can travel a speed of 500 km per hour.

3.17.3 Josephson Devices

Persistence of current in an insulator which is separated by an insulator, even in the absence of an applied voltage pairs of electrons moving through the potential barrier induce the superconducting current. This effect is known as Josephson effect.

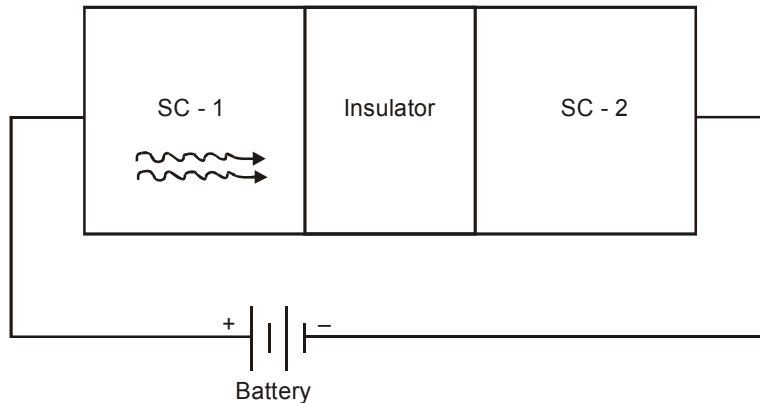


Fig. 3.40 Josephson effect

A thin layer of insulating material (10^{-5}Å) is placed in between two superconducting materials as shown in figure. When the voltage is applied across the superconductors, current start flowing between the superconductors.

The flowing current has both a.c and d.c components. The a.c current exists only up to which the external voltage is applied whereas the d.c current exist even after the removal of applied voltage. This effect is called Josephson effect.

SOLVED PROBLEMS

1. A paramagnetic material has a magnetic field intensity of 10^4 Am^{-1} . If the susceptibility of the material at room temperature is 3.7×10^{-5} . Calculate the magnetization and flux density in the material.

(AU - APRIL 2002, JUNE 2010, 2012)

Solution:

Given data:

$$\text{Magnetic field intensity } H = 10^4 \text{ Am}^{-1}$$

$$\text{Susceptibility } \chi = 3.7 \times 10^{-4}$$

$$1. \text{ Susceptibility } \chi = \frac{M}{H}$$

$$\begin{aligned} \therefore \text{Magnetization } M &= \chi H \\ &= 3.7 \times 10^{-3} \times 10^4 \\ &= 3.7 \times 10 \end{aligned}$$

$$\text{Magnetization } M = 37 \text{ A m}^{-1}$$

$$\begin{aligned} 2. \text{ Flux density } B &= \mu_0 (M + H) \\ &= 4\pi \times 10^{-7} \times (37 + 10^4) \\ &= 126179.4 \times 10^{-7} \\ &= 0.0126 \text{ Wb m}^{-2} \end{aligned}$$

$$\text{Magnetization } M = 37 \text{ A m}^{-1}$$

$$\text{Flux density } B = 0.0126 \text{ Wb m}^{-2}.$$

2. A magnetic material has a magnetization of 2300 A m^{-1} and produces a flux density of $0.00314 \text{ Wb m}^{-2}$. Calculate the magnetizing force and the relative permeability of the material.

Solution :

Given data:

$$\text{Magnetization } M = 2300 \text{ A m}^{-1}$$

$$\text{Flux density } B = 0.00314 \text{ Web m}^{-2}.$$

i) The magnetic flux density

$$B = \mu_0 (M + H)$$

$$\text{The magnetic force } H = \left[\frac{B}{\mu_0} - M \right]$$

$$= \frac{0.00314}{4\pi \times 10^{-7}} - 2300$$

$$H = 198.7326 \text{ A m}^{-1}$$

ii) Susceptibility $\chi = \frac{M}{H} = (\mu_r - 1)$

$$\therefore \text{Relative permeability } \mu_r = \frac{M}{H} + 1$$

$$= \frac{2300}{198.7326} + 1$$

$$\mu_r = 12.573$$

$$\text{Magnetic force } H = 198.7326 \text{ A m}^{-1}$$

$$\text{Relative permeability } \mu_r = 12.573$$

- 3. A paramagnetic material has FCC structure with a cubic edge of 2.5 Å. If the saturation value of magnetization is $1.8 \times 10^6 \text{ A m}^{-1}$, Calculate the magnetization contributed per atom in Bohr magnetrons.**

Solution:

Given data:

$$\text{The interatomic spacing } a = 2.5 \times 10^{-10} \text{ m}$$

$$\text{The magnetization } M = 1.8 \times 10^6 \text{ A m}^{-1}$$

The number of atoms present per unit volume

$$N = \frac{\text{Number of atoms present in a unit cell}}{\text{Volume of the unit cell}}$$
$$= \frac{2}{(2.5 \times 10^{-10})^3}$$

$$N = 1.28 \times 10^{29} \text{ m}^{-3}$$

Total magnetization $M = 1.8 \times 10^6 \text{ A m}^{-1}$

The magnetization produced per atom

$$= \frac{M}{N}$$
$$= \frac{1.8 \times 10^6}{1.28 \times 10^{29}}$$
$$= 1.4062 \times 10^{-23} \text{ A m}^{-2}$$

Bohr magneton $\mu_B = \frac{eh}{4\pi m}$

$$= \frac{1.6 \times 10^{-19} \times 6.625 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31}}$$

$$\mu_B = 9.27 \times 10^{-24} \text{ A m}^{-2}$$

$\therefore$ Magnetization produced per atom

$$M = \frac{1.40625 \times 10^{-23}}{9.27 \times 10^{-24}}$$

$$M = 1.519 \text{ Bohr magnetons}$$

The average magnetization per atom = 1.517 Bohr magnetons

4. The saturation magnetic induction of Nickel is 0.65 Wb m^{-2} . If the density of Nickel is 8906 kg m^{-3} and its atomic weight is 58.7 , calculate the magnetic moment of the Nickel atom in Bohr magnetron. (AU - DEC 2001)

Solution:

Given data:

Saturation magnetic induction of Nickel

$$B_a = 0.65 \text{ Wb m}^{-2}$$

The density of Nickel $\rho = 8906 \text{ Kg m}^{-3}$

Atomic weight of Nickel $W = 58.7$

The number of Nickel atoms $/\text{m}^3$

$$N = \frac{\rho \times \text{Avogadro number}}{\text{Atomic weight}}$$

$$= \frac{8906 \times 6.025 \times 10^{26}}{58.7}$$

$$N = 9.14 \times 10^{28} \text{ m}^{-3}$$

Saturation magnetization $B_a = N \mu_0 \mu_m$

$$\therefore \text{Magnetic moment } \mu_m = \frac{B_a}{N \mu_0}$$

$$= \frac{0.65}{9.14 \times 10^{28} \times 4 \times 3.14 \times 10^{-7}}$$

$$= 5.66 \times 10^{-24} \text{ A m}^2$$

$$= \frac{5.66 \times 10^{-24}}{9.29 \times 10^{-24}} \text{ Bohr magneton}$$

$$\mu_m = 0.61 \mu_B$$

Magnetic moment of Nickel $\mu_m = 0.61 \mu_B$

5. In a magnetic material the field strength is found to be 10^6 A m^{-1} . If the magnetic susceptibility of the material is 0.5×10^{-5} , calculate the intensity of magnetization and flux density in the material.

Solution:

Given data:

Magnetic field strength $H = 10^6 \text{ A m}^{-1}$

Susceptibility $\chi = 0.5 \times 10^{-5}$

i) Magnetization $M = \chi H$
 $= 10^6 \times 0.5 \times 10^{-5}$
 $M = 5 \text{ A m}^{-1}$

ii) Flux density $B = \mu_0 (M + H)$
 $= 4 \times 3.14 \times 10^{-7} (5 + 10^6)$
 $B = 1.257 \text{ Wb m}^{-2}$

Magnetization $M = 5 \text{ A m}^{-1}$

Flux density $B = 1.257 \text{ Wb m}^{-2}$

6. A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2 K.
 (AU - NOV 2002)

Solution:

Given data:

Critical temperature $T_c = 3.7 \text{ K}$

Critical field $H_c = 0.0306 \text{ Tesla}$

Temperature $T = 2 \text{ K}$

The critical magnetic field

$$H_c = H_0 \left[1 - \left[\frac{T^2}{T_c^2} \right] \right]$$

$$= 0.0306 \left[1 - \left[\frac{2}{3.7} \right]^2 \right] = 0.0216 \text{ Tesla}$$

The critical magnetic field $H_c = 0.0216 \text{ Tesla}$.

7. Calculate the critical current and current density for a wire of a lead having a diameter of 1 mm at 4.2 K. The critical temperature for lead is 7.18 K and $H_0 = 6.5 \times 10^4 \text{ A m}^{-1}$. (AU - MAY 2003, JUNE 2006)

Solution:

Given data:

Critical temperature $T_c = 7.18 \text{ K}$

Critical field $H_0 = 6.5 \times 10^4 \text{ A m}^{-1}$

Temperature $T = 4.2 \text{ K}$

Radius of the wire $r = 0.5 \times 10^{-3} \text{ m}$

$$\text{The critical magnetic field } H_c = H_0 \left[1 - \left[\frac{T^2}{T_c^2} \right] \right]$$

$$= 6.5 \times 10^4 \left[1 - \left[\frac{4.2}{7.18} \right]^2 \right]$$

$$= 4.276 \times 10^4 \text{ A m}^{-1}$$

i) Critical current $I_c = 2\pi r H_c$

$$= 2 \times 3.14 \times 0.5 \times 10^{-3} \times 4.276 \times 10^4$$

$$= 134.39 \text{ A}$$

ii) Critical density $J_c = \frac{I_c}{\pi r^2}$

$$= \frac{134.39}{3.15 \times (0.5 \times 10^{-3})^2}$$

$$= 1.71 \times 10^8 \text{ A m}^{-2}$$

Critical current $I_c = 134.39 \text{ A}$

Critical density $J_c = 1.71 \times 10^8 \text{ A m}^{-2}$

8. Prove that susceptibility of superconductor is -1 and relative permeability is zero.

Solution:

Given data:

We know, the induced magnetic field $B = \mu_0(M + H)$... (1)

In superconductor, $B = 0$

Therefore, $0 = \mu_0(M + H)$

Since $\mu_0 \neq 0$, $M = -H$

$$\frac{M}{H} = \chi = -1 \quad \dots (2)$$

Also, $\chi = \mu_r - 1$

$$-1 + 1 = \mu_r = 0$$

Therefore the susceptibility $\chi = -1$

and Relative permeability $\mu_r = 0$

9. Find the critical current which can pass through a long thin superconducting wire of aluminum of diameter 2 mm, the critical magnetic field for aluminum is $7.9 \times 10^3 \text{ A m}^{-1}$.

Solution:

Given data:

The critical magnetic field $H_c = 7.9 \times 10^3 \text{ A m}^{-1}$

$$\begin{aligned} \text{Radius } r &= \frac{\text{Diameter}}{2} \\ &= \frac{2}{2} = 1 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} \text{Critical current } I_c &= 2\pi r H_c \\ &= 2 \times 3.14 \times 1 \times 10^{-3} \times 7.9 \times 10^3 \text{ A m}^{-1} \end{aligned}$$

$$I_c = 49.65 \text{ A}$$

Critical current $I_c = 49.65 \text{ A}$

10. The superconducting transition temperature of Lead is 7.26 K. The initial field at 0 K is $64 \times 10^3 \text{ Amp m}^{-1}$. Calculate the critical field at 5 K.

(AU - JUNE 2012)

Solution:

Given data:

Critical temperature $T_c = 7.26 \text{ K}$

Critical Field $H_0 = 64 \times 10^3 \text{ Amp m}^{-1}$

Temperature $= 5 \text{ K}$

$$\begin{aligned}\text{The Critical Field } H_c &= H_0 \left[1 - \left[\frac{T}{T_c} \right]^2 \right] \\ &= 64 \times 10^3 \left[1 - \left[\frac{5}{7.26} \right]^2 \right] \\ &= 64 \times 10^3 \times 0.5257 \\ H_c &= 33.644 \times 10^3 \text{ Amp m}^{-1}\end{aligned}$$

11. A magnetic field of 2000 Amp m^{-1} is applied to a material which has a susceptibility of 1000. Calculate the (i) Intensity and (ii) Flux density.

(AU - JUNE 2009)

Solution:

Given data:

Magnetic Field $H = 2000 \text{ Amp m}^{-1}$

Susceptibility $\chi = 1000$

i) Magnetisation

$$\begin{aligned}M &= \chi H \\ &= 2000 \times 1000 \\ &= 2 \times 10^6 \text{ Amp m}^{-1}\end{aligned}$$

ii) Flux density

$$\begin{aligned}B &= \mu_0 (M + H) \\ &= 4 \times 3.14 \times 10^{-7} (2 \times 10^6 + 2000) \\ B &= 2.514 \text{ wbm}^{-2}\end{aligned}$$

- 12. The superconducting transition temperature of lead of 7.26 K. The initial field at 0 K is $64 \times 10^3 \text{ Amp m}^{-1}$. Calculate the critical field at 5 K.**

(AU - JULY 2010)

Solution:

Given data:

$$\text{Critical temperature } T_c = 7.26 \text{ K}$$

$$\text{Critical field } H_0 = 64 \times 10^3 \text{ Amp m}^{-1}$$

$$\text{Temperature } T = 5 \text{ K}$$

$$\begin{aligned} \text{The critical field } H_c &= H_0 [1 - (T/T_c)^2] \\ &= 64 \times 10^3 [1 - (5/7.26)^2] \end{aligned}$$

$$= 64 \times 10^3 \times 0.5257$$

$$H_c = 33.644 \times 10^3 \text{ Amp m}^{-1}$$

- 13. The magnetic field strength of Silicon is 1500 Amp m^{-1} . If the magnetic susceptibility is (-0.3×10^{-5}) , calculate the magnetization and flux density in Silicon.**

(AU - JUNE 2011, 2013)

Solution:

Given data:

$$\text{Magnetic susceptibility } \chi = -0.3 \times 10^{-5}$$

$$\text{Magnetic field strength } H = 1500 \text{ Amp m}^{-1}$$

$$\text{Magnetic susceptibility } \chi = \frac{I}{H}$$

$$I = \chi H$$

$$= -0.3 \times 10^{-5} \times 1500$$

$$= -4.5 \times 10^{-3} \text{ Amp m}^{-1}$$

We know

$$\mu = H + \chi$$

$$= 1 - 4.5 \times 10^{-3}$$

$$\mu = 0.999$$

$$\text{Flux density } B = \mu H$$

$$= 0.999 \times 1500$$

$$B = 1498.5 \text{ web m}^{-2}$$

- 14. Calculate the critical current which can flow through a long thin super conducting wire of diameter 1 mm. The critical magnetic field is $7.9 \times 10^3 \text{ Amp m}^{-1}$. (AU - MAY 2011)**

Soution:

Given data:

$$\text{Diameter of the wire } d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$\text{radius of the wire } r = \frac{d}{2} = \frac{1 \times 10^{-3}}{2} \text{ m}$$

$$\text{The critical magnetic field } H_c = 7.9 \times 10^3 \text{ Amp m}^{-1}$$

Critical current flowing through the wire

$$I_c = 2\pi r H_c$$

$$= 2 \times 3.14 \left(\frac{1 \times 10^{-3}}{2} \right) (7.9 \times 10^3)$$

$$I_c = 24.81 \text{ Amp}$$

SHORT QUESTIONS WITH ANSWERS

1. Define Magnetic Flux (ϕ).

The total number of number magnetic lines of force passing through a surface is known as magnetic flux.

Unit: Weber.

2. Define magnetic induction (B) or magnetic flux density.

It is the number of magnetic lines of force passing through unit area of cross section.

$$B = \frac{\phi}{A} \text{ Wb m}^{-2} \text{ (or) Tesla}$$

3. Define Magnetic field strength or Magnetizing field.

It is the force experienced by a unit North Pole placed at a given point in a magnetic field. The magnetic induction B due to the magnetic field of intensity H applied in vacuum is related by $B = \mu_0 H \text{ Amp m}^{-1}$.

4. Define intensity of magnetization (I)

Magnetization is the process of converting a non-magnetic material in to a magnetic material. It is also defined as the magnetic moment per unit volume.

$$I = m / V.$$

Unit : Web / m².

5. Define Magnetic permeability (μ).

It is the ratio of the magnetic induction (B) to the applied magnetic field intensity (H). $\mu = B / H$

Unit : Henry m⁻¹.

6. Define relative permeability (μ_r).

It is defined as the ratio of permeability of the medium to the permeability of the free space (μ_r). $\mu_r = \mu / \mu_0$.

7. Define magnetic susceptibility (χ).

It is defined as the ratio of intensity of magnetization (I) and intensity of magnetic field (H). $\chi = I / H$. The sign and magnitude of χ are used to determine the nature of the magnetic materials.

8. Define Bohr magneton (μ_B).*(AU - June 2010, Nov 2010)*

The orbital magnetic moment and the spin magnetic moment of an electron in an atom can be expressed in terms of atomic unit of magnetic moment called Bohr magneton.

$$1 \text{ Bohr magneton} = \frac{e\hbar}{2m} \Rightarrow \mu_B = 9.27 \times 10^{-24} \text{ Am}^2$$

9. How the magnetic materials are classified?*(AU - July 2009)*

The magnetic materials are classified into two categories.

The materials without permanent magnetic moment

Example: 1. Diamagnetic materials.

The materials with permanent magnetic moment.

Example: 1. Paramagnetic materials
2. Ferromagnetic materials
3. Anti-Ferromagnetic materials
4. Ferrimagnetic materials.

10. Define Diamagnetic Materials.

In a diamagnetic material the electron orbits are randomly oriented and the orbital magnetic moments get cancelled. Similarly, all the spin moments are paired i.e., having even number of electrons. Therefore, the electrons are spinning in two opposite directions and hence the net magnetic moment is zero. These materials are called as diamagnetic materials.

Example: Gold, germanium, silicon, antimony, bismuth, silver, lead, copper, hydrogen, Water and alcohol.

11. Define Paramagnetic Materials.

Para magnetism is due to the presence of few unpaired electrons which gives rise to the spin magnetic moment. In the absence of external magnetic field, the magnetic moments (dipoles) are randomly oriented and possess very less magnetization in it. When an external magnetic field is applied to paramagnetic material, the magnetic moments align themselves along the field direction and the material is said to be magnetized.

Example: Platinum, CuSO_4 , MnSO_4 Palladium, Chromium, Aluminum, etc

12. Define Ferromagnetic Materials.

(AU - June 2010, Dec 2009)

Ferromagnetism is due to the presence of more unpaired electrons. Even in the absence of external field, the magnetic moments align parallel to each other. So that it has large magnetism. This is called spontaneous magnetization.

Example: Nickel, iron, Cobalt, Steel, etc.

13. What is the Effect of magnetic field on Ferromagnetic materials?

(AU - June 2012)

If a small external magnetic field is applied the magnetic moments align in the field direction and become very strong magnets.



14. What are the properties of Ferromagnetic Materials?

1. All the magnetic lines of force pass through the material.
2. Its susceptibility is **high positive** and it is given by $X = \frac{C}{T - \theta}$
3. The permeability is very much greater than one.
4. They have enormous permanent dipole moment.
5. When the temperature is greater than the Curie temperature, the Ferromagnetic material becomes paramagnetic material.
6. The ferromagnetic material has equal magnitude dipole lying parallel to each other.

Examples: Nickel, iron, Cobalt, Steel, etc.

15. What are magnetic domains?

A ferromagnetic material is divided into a large number of small regions called domains. (0.1 to 1 mm² of area). Each direction is spontaneously magnetized. The direction of magnetization varies from domain to domain and the net magnetization is zero in the absence external magnetic field. The boundary line which separates two domains is called domain wall or Block wall.

16. Define Exchange Energy.

The energy which makes the adjacent dipoles to align themselves is known as exchange energy. It is also called as magnetic field energy or magneto static energy. It arises from interaction of electron spins and it depends upon the inter atomic distance.

17. What is Anisotropy energy?

Crystals are anisotropic, the energy arises from the difference of energy required for magnetization along any two different directions in a single crystal. These are two directions of magnetization.

1. Easy direction.

2. Hard direction.

In easy direction of magnetization, weak field can be applied and in hard direction of magnetization, strong field should be applied.

The excess energy required to magnetize a specimen in particular direction over that required to magnetize it along the easy direction is called the crystalline anisotropy energy.

18. What is Domain wall energy (or) Bloch wall energy?

It is the sum of the exchange and anisotropy energy in the domain wall. Thickness of the wall is approximately $1000 \text{ \AA}$.

19. What is Magnetostriction energy?

When the domain is magnetized in different directions, they will either expand or shrink. i.e., Change in dimension when it is magnetized. The energy produced in this effect is called Magnetostriction energy. It is the energy due to the mechanical stresses generated by domain rotations.

20. Give the experimental evidence for Domain Structure?

An experimental evidence for domain structure was given by **BITTER** and is called Bitter powder pattern. In this method, a drop of colloidal suspension of finely divided ferromagnetic powder is allowed to spread over the surface of the ferromagnetic material. It is found that through the microscope, the colloidal particles are collected along the domain boundaries which shows the existence of domain structure.

21. Define Hysteresis.

Hysteresis means “Lagging” i.e., The Lagging of intensity of magnetization (I) behind the intensity of magnetic field (H) which is applied is called Hysteresis.

22. What is meant by Hysteresis Loss?

When the specimen is taken through a cycle of magnetization, there is loss of energy in the form of heat. This is known as Hysteresis Loss.

23. Define Retentivity and Coercivity. (AU - Dec 2010)

During the process of demagnetization, the material retains some amount of magnetism, even though when intensity of magnetic field is zero. It is known as Retentivity or residual magnetism.

The amount of intensity of magnetic field applied in the reverse direction to remove the retentivity is known as **coercivity or coercive force**.

24. What are reversible and irreversible domains?

When a magnetic field is applied, the domain walls are displaced and gives rise to small value of magnetization. Now if the field is removed, the domains return to its original state and are known as reversible domains.

When the field is removed certain domain boundaries do not come back to the original position due to the domain wall movement to a very large distance. These domains are called irreversible domains.

25. What are soft magnetic materials? Give its properties? (AU - June 2009)

Soft Magnetic materials: Materials which are easy to magnetize and demagnetize are called soft magnetic materials.

These materials do not retain the alignment of magnetic domains after the removal of external magnetic field.

Properties

1. They have **high** Permeability
2. The storing of magnetic energy is **less**
3. Retentivity and Coercivity values are **small**
4. They have **low** hysteresis loss

26. What are hard magnetic materials? Give its properties?

Hard Magnetic materials: Materials which retain their magnetism and are difficult to demagnetize are called hard magnetic materials.

These materials retain permanently the alignment of magnetic domains even after the removal of external magnetic field.

Properties

1. They have **low** Permeability
2. The storing of magnetic energy is **more**
3. Retentivity and Coercivity values are **high**
4. They have **high** hysteresis loss

27. Define Energy product.

(AU - Dec 2009)

The product of residual magnetic induction (B_r) and coercivity (H_c) is called energy product ($B_r \times H_c$). The value of energy product is very high for permanent magnets.

Importance:

- 1) It is used to design powerful permanent magnets
- 2) It is used to maximum amount of energy stored in the specimen

28. Distinguish between Soft and Hard magnetic material. *(AU - June 2009)*

<i>S. No</i>	<i>Hard Magnetic Materials</i>	<i>Soft Magnetic Materials</i>
1.	Cannot be easily magnetized	Can be easily magnetized.
2.	It can be produced by heating and sudden cooling	It can be produced by heating and slow cooling.
3.	Domain wall does not move easily and require large value of H for magnetization.	Domain wall move easily and requires small value of H for magnetization.
4.	Hysteresis loop area is large Susceptibility and Permeability values are small.	Hysteresis loop area is small Susceptibility and Permeability values are high.
5.	Retentivity and Coercivity are large	Retentivity and Coercivity are small.
6.	High eddy current loss	Low eddy current loss
7.	Impurities and defects will be more	No impurities and defects
8.	Examples: Alnico, Chromium steel, tungsten steel, carbon steel.	Examples: Iron-silicon alloy, Ferrous nickel alloy, Ferrites Garnets.
9.	Uses: Permanent magnets, DC magnets.	Uses: Electro magnets, computer data storage. Transformer core.

29. Define Anti ferromagnetic materials.**(AU - June 2010)**

Magnetic materials in which, the spins are aligned in anti-parallel manner due to un favorable exchange interaction among them resulting in zero magnetic moment are called as Anti ferromagnetic materials.

30. What are Ferrimagnetic materials or Ferrites?

Ferromagnetic materials are much similar to ferromagnetic materials in which the magnetic dipoles are aligned anti-parallel with unequal magnitudes. If small value of magnetic field is applied, it will produce the large value of magnetization.

31. What are the Applications of Ferrites?**(AU - May 2004, 2011)**

1. Ferrites are used to produce ultrasonic wave by Magnetostriction principle.
2. Ferrites are used in audio and video transformers.
3. Ferrites rods are used in radio receivers to increase the sensitivity.
4. Ferrites are widely used in non-reciprocal microwave devices such as gyrator, circulator and Isolator.
5. They are also used for power limiting and harmonic generation.
6. Ferrites are used in parametric amplifiers so that the input can be amplified with low noise.
7. They are used in computers and data processing circuits.

32. What Magnetic Bubbles?**(AU - May 2005, April 2002)**

Magnetic bubble is direct access storage medium. Magnetic bubbles are soft magnetic materials with magnetic domains of a few micrometers in diameter. These bubbles are the electrical analogue of the magnetic disk memories used in computers. The magnetic disk in the hard disk memory is moved mechanically where as the bubbles in a bubble memory device are moved electronically at very high speeds.

33. What is Superconductivity?**(AU -Nov 2003, April 2002, Dec 2009)**

The ability of certain metals and alloys exhibit almost zero electrical resistivity when they are cooled to low temperature is known as superconductivity. (ie., maximum conductivity with zero resistance at zero Kelvin)

34. Define Critical Temperature.

(AU - June 2009)

Critical temperature (or) Temperature (T_C) is defined as the temperature at which the resistivity falls to zero is called transition as temperature or Critical temperature. The temperature at which the normal conductor becomes a superconductor is known as critical temperature (T_C). Below T_C , the material (Tin) is in the superconducting state and above T_C , it is in the normal state.

35. State BCS Theory?

(Bardeen, Cooper and Schrieffer theory). It states that an electron pair called Cooper pair is formed at low temperature by overcoming the repulsive force. The Cooper pair moves without scattering (without any resistance) in the lattice structure. Now the material becomes a superconductor.

This is also called as electron-phonon-electron interaction. These two electrons have equal and opposite momentum and spins.

This BCS theory is suitable only for low temperature superconductors.

36. Define Cooper pair.

(AU - June 2012)

The electron pairs called Cooper pairs are formed at low temperature by overcoming the repulsive forces. The Cooper pairs move without scattering (without any resistance) in the lattice structure. Now the material becomes superconductor. This is also called as electron-phonon-electron interaction. These two electrons have equal and opposite momentum and spins.

37. What is coherence length?

It is defined as the distance up to which two electrons combine to form a Cooper pair.

38. What are the important properties of superconductors?

1. **Zero Electrical Resistance :** The electrical resistivity drops to zero at the transition temperature (T_C). Dr. Collins observed that the current in a superconducting ring is unchanged for 2½ years.
2. **Effect of magnetic field :** Below T_C , the superconducting property disappears when a strong magnetic field is applied. i.e., it returns to normal state.
3. **Diamagnetic property :** The superconductor is a perfect diamagnetic.

4. **Effect of heavy Current** : The superconducting property disappears when a heavy current flows, since current flow will set up a magnetic field.
5. **Isotope effect** : The presence of isotopes in superconductors changes the transition temperature of the superconductors.
6. **Thermal properties** : Entropy and specific heat decreases at transition temperature and the thermoelectric effect disappear in the superconducting state.
7. **Effect of Impurities** : The impurities change the magnetic properties of the superconductors.

39. Define Critical magnetic field.

A very strong magnetic field applied to superconducting material it disappears superconducting property this is called as critical magnetic field.

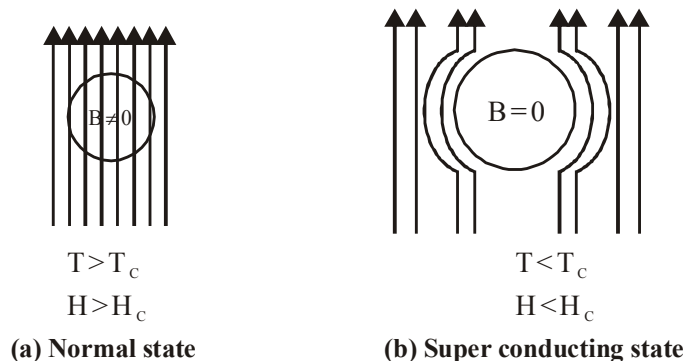
$$H_C = H_0 \left[1 - \frac{T^2}{T_c^2} \right] \quad H_0 \text{ Where } = \text{Critical field at } 0 \text{ K}$$

When the temperature of the material increases, the value of critical magnetic field decreases.

40. What is Meissner effect? (or) Define Diamagnetic property.

(AU., May 2008, June 2006, 2010, Nov 2010)

When the superconducting material is subjected to a uniform magnetic field, under the condition $T \leq T_c$ and $H \leq H_c$, the magnetic flux lines are excluded from the material. Thus the material exhibits perfect diamagnetism. This phenomenon is called as Meissner effect.



41. What happens in a superconductor when a heavy Current flows?

The superconducting property disappears when a heavy current flows, since this current flow will set up a heavy magnetic field. According to the Silsbee's rule, for a superconducting wire $I_c = 2\pi r H_c$.

42. What is Isotope effect?

(AU - June 2006, Dec 2010)

The presence of isotopes in superconductors change the transition temperature of the superconductor. The transition temperature is found to be inversely proportional to the square root of the atomic weight of the isotope.

$$T_c \propto \frac{1}{\sqrt{M}}$$

$$T_c = \frac{\text{const} \tan t}{\sqrt{M}}$$

$$\sqrt{M} \times T_c = \text{Constant}$$

The transition temperature of the heavier isotope is less than that of the lighter isotopes.

43. How the superconductors are classified? or what are the types of superconductors.

Based on the value of H_c we have,

- 1) Type I (or) Soft superconductors
- 2) Type II (or) Hard superconductors

Based on the value of T_c we have,

- 1) High temperature superconductors
- 2) Low temperature superconductors

44. What is Type I superconductors? Give its characteristics.

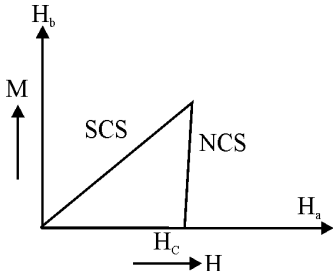
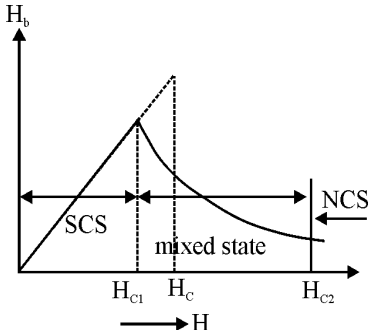
In type I superconductor, the magnetic field is completely excluded from the material below the critical magnetic field H_c and the material loses its superconducting property abruptly at H_c .

Characteristics

1. They exhibit complete Meissner Effect.
2. They have only one critical magnetic field value.
3. Below the material behaves as superconductor and above the material behaves as normal conductor.
4. These are called as Soft superconductors.

45. Distinguish between Type I and Type II Superconductors

(AU - Dec 2005, April 2003)

S.N	Type I (or) Soft superconductors	Type II (or) Hard superconductors
1.	It exhibits complete Meissner Effect.	It does not exhibit a complete Meissner Effect.
2.	They are completely diamagnetic.	They are not completely diamagnetic.
3.	Ex : Tin, Lead, Mercury, etc.,	Ex : Nb-Zr, Nb- Ti, Nb-Sn, Va- Ga, etc.,
4.	 $H_a \rightarrow$ Applied magnetic field $H_b \rightarrow$ Induced magnetic field	
5.	The value of H_c is always too low. It is about 0.1 tesla.	The values of H_c is high and it is about 30 tesla.
6.	They have only one critical magnetic field value.	They have two critical magnetic field values. Lower critical magnetic field [H_{c1}] and Higher critical magnetic field [H_{c2}].
7.	No mixed state exists.	Mixed state is present.
8.	These are called Soft superconductors.	These are called Hard superconductors.
9.	It requires low magnetic field to destroy the superconductivity.	It requires large magnetic field to destroy the superconductivity.
10.	It loses its magnetization abruptly.	It loses its magnetization gradually.

46. What is Type II superconductors? Give its characteristics.

In type II superconductor, the magnetic field is excluded from the material and the material loses its superconducting property gradually rather than abruptly.

Characteristics

1. They do not exhibit a complete Meissner Effect.
2. They have two critical magnetic field values. Lower critical magnetic field [H_{C1}] and Higher critical magnetic field [H_{C2}].
3. Below H_{C1} the material behaves as superconductor and above the material behaves as normal conductor. The region in between [H_{C1}] and [H_{C2}] is called mixed state or vortex region.
4. These are called as Hard superconductors.

47. What are Low T_c and High T_c superconductors?

Low T_c Superconductors

1. The superconductors having the critical temperature less than 20 K are known as low T_c Superconductors or elemental superconductors.
2. The Superconductors by BCS theory.
3. It is explained by BCS theory.
4. It is not so useful due to its low temperature maintenance.
5. It is called as N-type superconductor.

High T_c Superconductors

1. The superconductors having the critical temperature greater than 100 K are known as high T_c Superconductors or ceramic or oxide superconductors.
2. The Superconductors is due to hole states.
3. It is explained by RVB theory proposed by Anderson.
4. It is very useful for commercial and engineering purposes.
5. It is called as P-type superconductor.

48. What are the characteristics of high temperature [high T_c] superconductors?

1. High Superconductors have high temperatures.
2. They have a modified perovskite crystal structure.
3. Superconducting state is direction dependent.
4. These are oxides of copper with other elements.
5. These are reactive, brittle, and cannot be easily modified or joined.
6. For high superconductors, liquid Nitrogen is used instead of liquid helium.

49. What are the Applications of Superconductors?

1. Electric generators can be made by using superconductors with smaller size, less weight and low energy consumption.
2. Superconductors can be used for the transmission of power over very long distances.
3. Superconductors transformers are available.
4. Superconductors can be used in switching Devices.
5. The superconductors can be used in sensitive electrical instruments.
6. It can be used as a memory or storage element in computers.
7. These are used to design Cryotron, Maglev, Josephson Devices and SQUID.
8. DC superconducting motors are used in ship propulsion and in large mills.
9. Superconducting magnetic field may be used to launch satellite into orbit directly from the earth without use of rockets.
10. Ore separation can be done by using machines made of superconducting magnets.
11. These are used in NMR (Nuclear Magnetic Resonance) imaging equipments which is used for scanning purposes.
12. Superconductors are used for the detection of brain tumor, defective cells, etc.,
13. Superconducting solenoids are used in magneto hydrodynamic power generation to maintain the plasma in the body.

50. What is Cryotron?

1. It is a magnetically operated current switch.
2. The superconducting property disappear when the magnetic field is greater than critical field (H_c).
3. It consists of a superconducting material [A] and it is surrounded by an another super conducting coil of wire [B]. When the critical magnetic field of wire B exceed or less than that of superconducting material A, the current in A can be controlled by the current in the material B.

51. Define Magnetic levitation?

(AU - May 2008)

When a magnet is placed over a superconductor, the magnet floats, this phenomenon is known as magnetic levitation. This is based on diamagnetic property of a superconductor. A diamagnetic material rejects the magnetic flux lines.

52. What is Maglev?

It is a magnetic levitated train. Electromagnetic induction principle is used here, this train cannot move over the rail. Instead it floats above the rails, so that it move faster with speed of 500 Km/hr without any frictional loss. It has two superconducting magnet on each side of the train and there is guiding system consisting of 'S' shaped coils on each side. Due to actions of these magnets the train moves faster thro' levitation.

53. What is SQUID?

(AU - June 2006)

SQUID stands for Superconductors Quantum Interference Device. It is a double junction quantum interferometer formed from two Josephson junctions mounted on a superconducting ring. SQUID is based on the flux quantization in a superconducting ring. The total magnetic flux passing through the ring is quantized. It is using to detect very minute magnetic field of the order of 10^{-14} tesla.

PART B – QUESTIONS

1. Write a note on magnetic storage. *(AU - April 2002)*
2. Explain in detail about the magnetic memories. *(AU - April 2002)*
3.
 - i) What is ferromagnetism?
 - ii) Explain the reason for the formation of domain structure in ferromagnetic material and how the hysteresis curve is explained on the basis of domain theory? *(AU - April 2002, June 2005)*
4.
 - i) Classify the magnetic materials on the basis of their spin?
 - ii) What are ferromagnetic domains? Explain. *(AU - Nov. 2002, May 2003)*
5. Explain the domain theory of ferromagnetism. Using that how will you explain the properties of ferromagnetic materials?
6.
 - i) What are ferromagnetic domains?
 - ii) Draw a B-H curve for a ferromagnetic material and identify retentive and coercive fields on the curve.
 - iii) What is energy loss per cycle?
 - iv) What are ferrites? *(AU - Dec 2003)*
7. What are hard and soft magnetic materials? Compare their properties. Give some examples.
8. Distinguish between dia, para, ferro and ferromagnetic material. Mention their properties and applications.

9. Explain the different structure of ferrites with suitable diagrams.
10. Write a short note on the following
 - i) Anisotropy energy ii) Magnetostrictive energy
 - iii) Domain wall energy iv) Exchange energy.
11. i) Explain ferromagnetism based on domain theory.
 ii) What are the four types energy involved in domain growth? Explain.
12. Discuss the domain structure in ferromagnetic materials. *(AU - Nov. 2003)*
13. Discuss the properties and applications of superconductors? *(AU - June 2006)*
14. i) Distinguish between Type –I and Type – II Superconductors.
 ii) Describe high temperature superconductors. *(AU - June 2006)*
15. i) What is super conductivity?
 ii) Mention any four properties of superconductors.
 iii) Explain the effect of isotopes in superconductors.
 iv) Explain type-I and type –II superconductors.
 v) Mention tow applications. *(AU - May 2003)*
16. i) Explain Meissner effect and magnetic levitation.
 ii) Discuss the applications of superconductors.
17. Mention the applications of superconductors. *(AU - April 2002, Nov.2007)*
18. Discuss type-I and type-II superconductors. *(AU - April 2002)*
19. Explain superconducting phenomena. What are its properties? What are its Applications? *(AU - May 2003)*
20. Distinguish type-I and type-II superconductors Write a note on high temperature superconductors.
21. Write an essay on superconducting materials and its applications. What are the new developments?
22. Explain the characteristics of superconducting materials. *(AU - Nov. 2007)*
23. Write a short note on
 - i) Meissner effect ii) Josephson effect
 - iii) Magnetic levitation iv) SQUID
24. Based on tunneling explain the A.C. and D.C. Josephson effects and its applications in the superconductors.

ASSIGNMENT PROBLEMS

1. The saturation value of magnetization of iron is $1.76 \times 10^6 \text{ A m}^{-1}$. Iron has body centered cubic structure with an elementary edge of $2.86 \text{ \AA}$. Calculate the average number of Bohr magnetons contributed to the magnetization per atom.

(Ans: 2.2 Bohr magneton per atom)

2. The magnetic field intensity of a ferric oxide piece is 10^6 A m^{-1} . If the susceptibility of the material at room temperature is 10.5×10^{-3} , calculate the flux density and magnetization of the material.

(Ans: $B = 1.259 \text{ T}$ and $M = 1500 \text{ A m}^{-1}$)

3. A magnetic material has a magnetization of 3000 A m^{-1} and flux density of 0.044 Wb m^{-2} . Calculate the magnetic force and the relative permeability of the material.

(Ans: $M = 203$ and $\mu_r = 17.26$)

4. Calculate the magnetic field in the lead at 5 K , if its critical magnetic field at 0 K $H_0 = 8 \times 10^5 \text{ A m}^{-1}$, and transition temperature $T_c = 7.26 \text{ K}$

(Ans: $4.2 \times 10^5 \text{ A m}^{-1}$)

5. The critical temperature T_c for mercury with isotopic mass 199.5 is 4.185 K . Calculate its critical Temperature, when its isotopic mass changes to 203.4 .

(Ans: 4.139 K)

6. Calculate the critical current which can flow through a long thin superconducting wire of aluminum of diameter 1 mm . The critical magnetic field for aluminum is $7.9 \times 10^3 \text{ A m}^{-1}$.